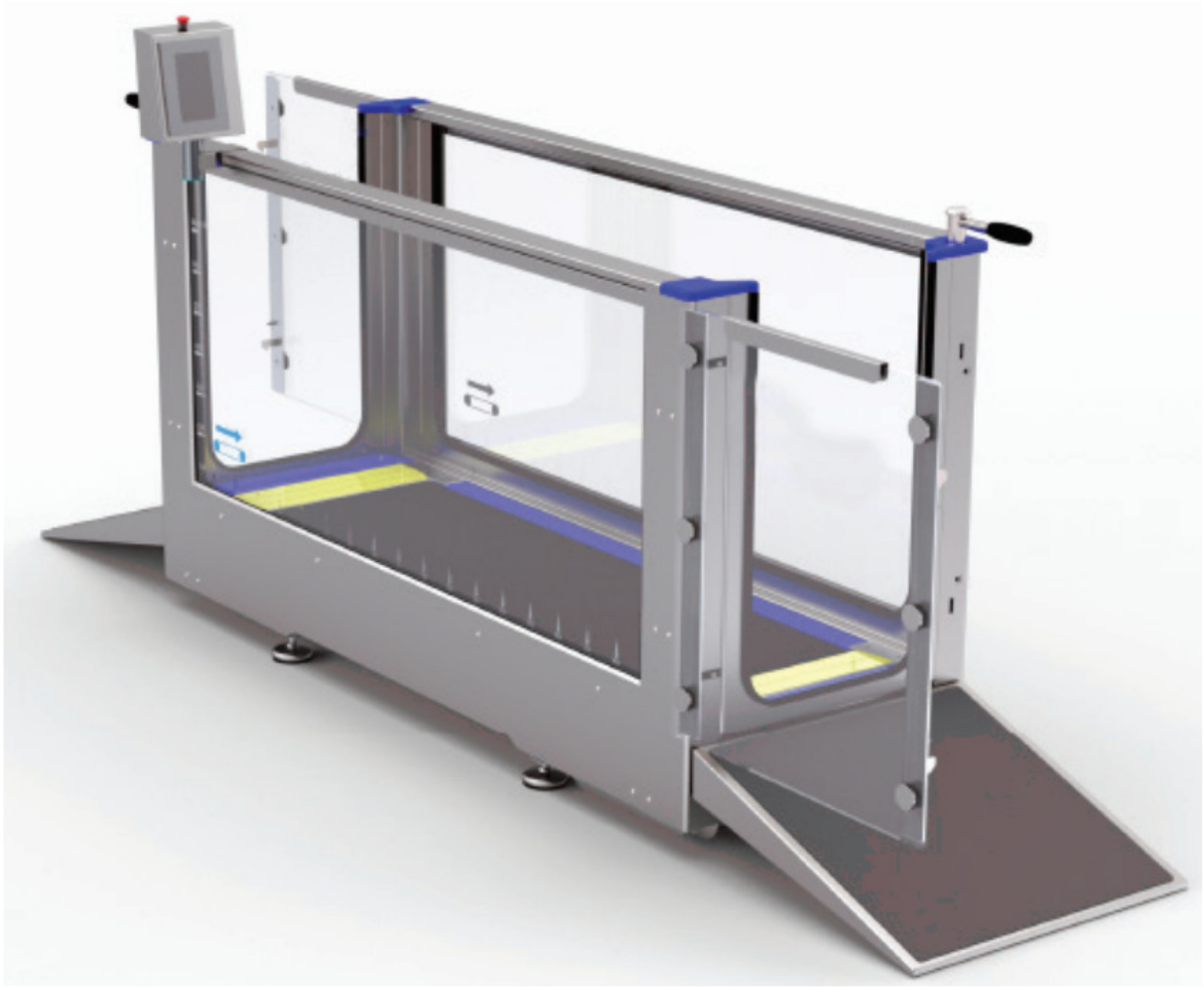
HYDROWORX

YOUR POTENTIAL ◆ ACHIEVED

Operations Manual

K900 Series



FOREWORD

Care has been taken with the design of this product to ensure that it is safe. However, misuse can result in injury to persons, damage to the equipment or serious injury. Therefore, it is very important that the instructions in this manual and on the product are observed during transportation, commissioning, operation, maintenance and disposal.

DOCUMENTATION SUPPLIED

Description	Document Number	Rev
Hydroworx System		**A**
Typical Layout	K900 - LAYOUT	**C**
Hydraulic Circuit	965/015000-HYDRAULIC CIRCUIT	**B**
Controls line drawing	50-CP0717-UK	**A**
Electrical Circuit EURO	K900 IEC Manual Schematics (EURO)	**V10**
Controls line drawing	50-CP717-UL	**A**
Electrical Circuit USA	K900 UL Manual Schematics (USA)	**-**
Appendix A - Transfer pump manual	pump_man_euroswim-271--uid672240	**-**
Appendix B - Circulation pump manual	pump_man_mini-280--uid672262	**-**
Appendix B-60Hz - Circulation pump manual	Pump Silen I	**-**
Appendix C - heater	Elecro FLOW LINE heater	**-**
Appendix C-60Hz - heater	Coates heater	**-**
Appendix D - UV Sanitizer manual	Certikin UV CUV-15V4	**-**
Appendix E - Filter	Brasil Max Filters English	**-**
Appendix F	pH_Reducer	
Appendix G	pH_Increaser	**-**
Appendix H - Bromine granules	Bromine data sheet	**-**
Quick start guide		**A**
Backwash & rinse guide		**A**
Water quality guide	K900 -water-quality-guide	**A**
Water quality sheet	K900 water-quality-sheet	**A**
Declaration of Conformity		
Warranty document		

CONTENTS

Foreword....2

Documentation Supplied....3

Contents....4

Specifications....6

Warranty Document....7

1. Introduction....8
   - 1.1 The components....8
   - 1.2 The Treadmill chamber....9
   - 1.3 The control system....11
   - 1.4 Water tank....13
   - 1.5 Plant pack....13
   - 1.6 The water tank and plant pack....13
   - 1.7 Water tank, plant pack and chamber....14
   - 1.8 Access ramps....14

2. K900 Users & Safety  15
   - 2.1 Responsibilities....15
   - 2.2 Operator....16
   - 2.3 Supervisors....16
   - 2.4 Emergency stop function....16
   - 2.5 Water temperature....17
   - 2.6 Protection for the glass panels....17
   - 2.7 Protection from electric shock....17
   - 2.8 Environment....17``
   - 2.9 Chamber drain time....17
   - 2.10 Maintenance personnel....17
   - 2.11 Installation personnel....18

3. Operation & Control System....19
   - 3.1 Starting the K900....19
   - 3.2 Door latch operation....19
   - 3.3 Switching On/Off....20
   - 3.4 Water temperature....23
   - 3.5 Touch screen operation....24
   - 3.6 General screen features....25
   - 3.7 Touch screen settings and administration....31

4. Cleaning....36
   - 4.1 Personal protective equipment (PPE) to be used when cleaning....36
   - 4.2 Materials used in construction....36
   - 4.3 Cleaning materials....36
   - 4.4 Daily cleaning procedure....37
   - 4.5 Weekly cleaning procedure....37
   - 4.6 Cleaning chamber glass and doors....37
   - 4.7 Cleaning underneath the treadmill....37
   - 4.8 Cleaning the sand filter....39
   - 4.9 Cleaning the pump filter baskets....41
   - 4.10 Changing the water in the water storage tank....42
   - 4.11 Dumping contaminated water from treadmill chamber....42

5. Routine Inspection & Maintenance....44
   - 5.1 Maintenance personnel - safety....44
   - 5.2 Maintenance schedules (Water management)....44
   - 5.3 Maintenance procedures....44
   - 5.4 Routine inspection and maintenance of plant pack components....45
   - 5.5 Routine inspection and maintenance of chamber components....47
   - 5.6 The water treatment system....48
   - 5.7 Bromine systems....49

5.8 Chlorine systems 50
5.9 Microbial tests 50
5.10 Spares/Parts list 51
5.11 USA Spares/Parts list 51
5.12 Spare parts 51

6. Troubleshooting 52

6.1 Fault information messages 52
6.2 Troubleshooting 52
6.3 Servicing information 53

7. Installation & Commissioning 54

7.1 Transportation and packaging 54
7.2 Safety information 55
7.3 Plant pack and water storage installation 56
7.4 Siting the plant pack 57
7.5 Exercise chamber installation 57
7.6 Siting the treadmill chamber 57
7.7 Water drain 57
7.8 Other electrical equipment in the vicinity of the exercise chamber 57
7.9 Ventilation 58
7.10 Cleaning 58

8. Decommissioning & Disposal 59

8.1 Lifespan of the K900 59
8.2 Decommissioning procedure 59
8.3 Disposal 59

SPECIFICATIONS

Type USA	965/015200
Type Europe	965/015100
Date of first supply	September 2022
Control software version	K900
Power supply USA	220Vac, 25 amp, 60Hz, single phase with 5mA RCD
Power supply Europe	220-240Vac, 16 amp, 50Hz, single phase
Drive motor	AC 200v 0.75kW
Control circuit voltage	24vDC
Control type	touch screen, HMI/PLC control panel, emergency-stops on main control panel and operator panel.
IP ratings, submerged components	IP68
IP Rating of plant pack electrical equipment	IP65
Voltage of plant pack electrical equipment	230v
Volume of water in chamber at 600mm depth	277 Gal. (1050 Litre)
Volume of water in the whole system	396 Gal. (1500 Litre)
Chamber mass	860 Lbs (390 kg) Dry
Plant pack mass	275 Lbs (125 kg) Dry
Water tank mass	214 Lbs (97kg) Dry
Water temperature range	Ambient to 104 ℉ (40 ℃)
Treadmill speed forward	0.1 to 6.8 mph
Treadmill speed reverse	0.1 to 1.2 mph
Treadmill chamber max fill depth	24” (600mm)
Treadmill incline	Level to 3.63 degrees
Manual issue	B

WARRANTY

Any damage caused by misuse is not covered under warranty, including any symptoms of incorrect water chemistry.

A water quality sheet is supplied with this manual, which must be kept up to date and available during our service visits.

To keep the system operating efficiently it should be serviced by HydroWorx Service after 1500 hours use or 12 months whichever is the sooner.

HAZARD

Incorrect water chemistry can be harmful to animals, users and the equipment. Always refer to the manufacturer’s procedures before handling chemicals.

Please contact our service team for advice on water chemistry and supply of consumables.

WARRANTY DOCUMENT

Canine HydroWorx Treadmill

Thank you for choosing HydroWorx the leading manufacturer and supplier of canine hydrotherapy treadmills.

Your new hydrotherapy treadmill is a high quality product constructed of quality materials to stringent standards.

To ensure longevity and trouble free operation, before using the equipment, please study the **Instruction manual** to gain a thorough understanding of the equipment, it's operation and maintenance requirements.

In the unlikely event a fault occurs please take note of the screen messages and refer to the Troubleshooting section of the operating instructions.

In the event a problem occurs that is not described in the operating instructions please revert to your dealer for further advice.

Following dealer intervention, if parts covered under the warranty are found to be faulty during the warranty period due to manufacturing defect these would be supplied by your HydroWorx distributor.
The warranty covers the cost of parts only. The cost of fitting, if required, would be advised by your distributor.

Parts replaced under warranty continue to be covered by the remaining original warranty period.
Spare parts must be sourced from an authorised distributor or HydroWorx.

The warranty covers replacement parts only and activated upon commissioning, or 3 months from date of despatch whichever is the sooner.
Treadmill drive motor and Controls – 24 months.
All other components apart from wear items – 12 months.
Wear items not covered by warranty - Door seals - Treadmill belt.

In the case of warranty, replacement parts should be ordered from HydroWorx Service, the cost of which would be credited should the parts prove to be faulty.

The warranty is invalidated if any modifications or alterations are made to the machine or working parts by none appointed HydroWorx distributors

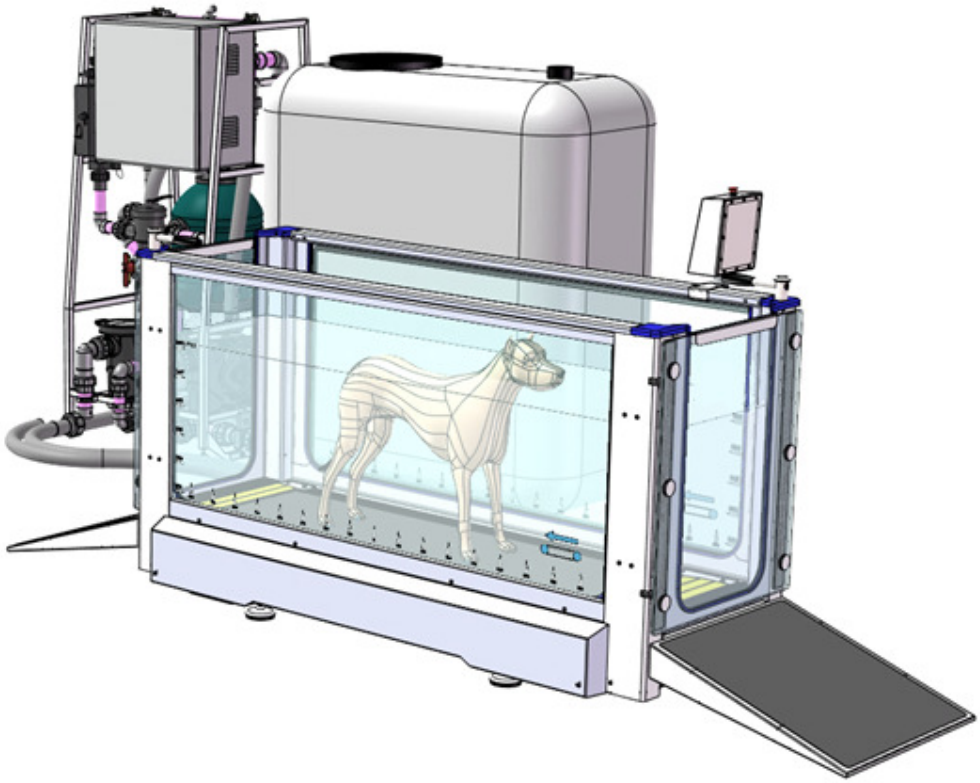
Warranty is not transferable.
Exclusions to this warranty include:
Damage caused by misuse.
Damage caused by neglect.
Deliberate damage.
Accidental damage.

Parts damaged thorough misuse or lack of service are not covered by warranty.
Please have your system serviced at the recommended intervals to validate warranty.

1. INTRODUCTION

The K900 treadmills are based on well established, hydrotherapy treatments, allowing animals to exercise in a supported and controlled environment. The warm water used in the K900 chamber helps promote blood circulation, whilst helping to relax recuperating muscles. This facilitates an early return to exercise for animal and reduces rehabilitation time.

The K900 treadmill has been designed with continuous veterinary input and incorporates many unique features that has made this the market leader and the first choice for veterinary surgeries worldwide.



1.1 The components

The system is comprised of four main component parts:

- Treadmill chamber
- Control system
- Water tank
- Plant pack (water management)

1.2 The Treadmill chamber

The stainless-steel chamber includes dual entry/exit doors with access ramps and all round glass. The ability to view an animal during therapy allows the professional operator to easily control and monitor the rehabilitation process.

Doors are secured via a unique top operation locking mechanism. This allows for the operator to open and close the door easily from inside and outside of the treadmill chamber.

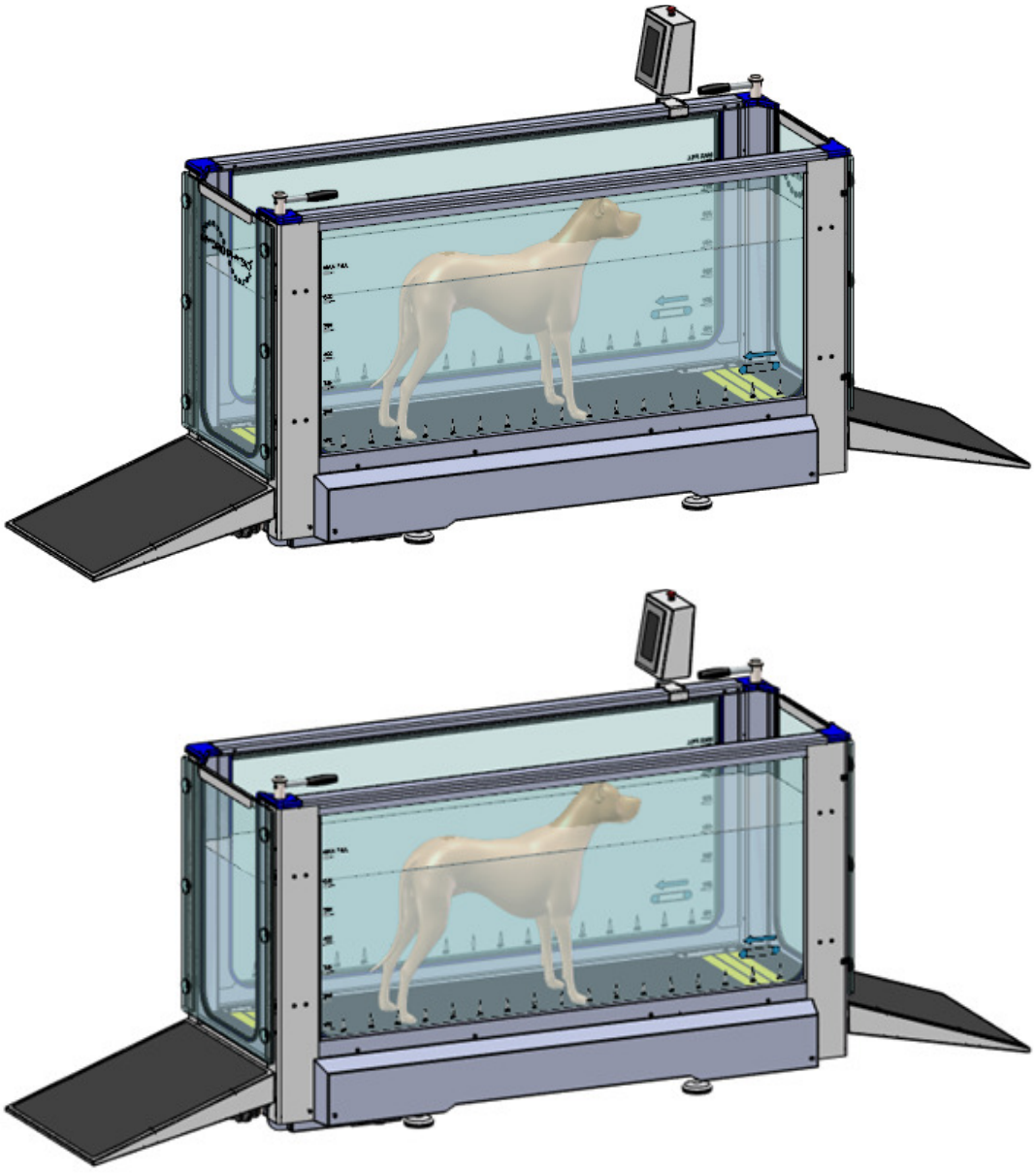


Figure 1-1

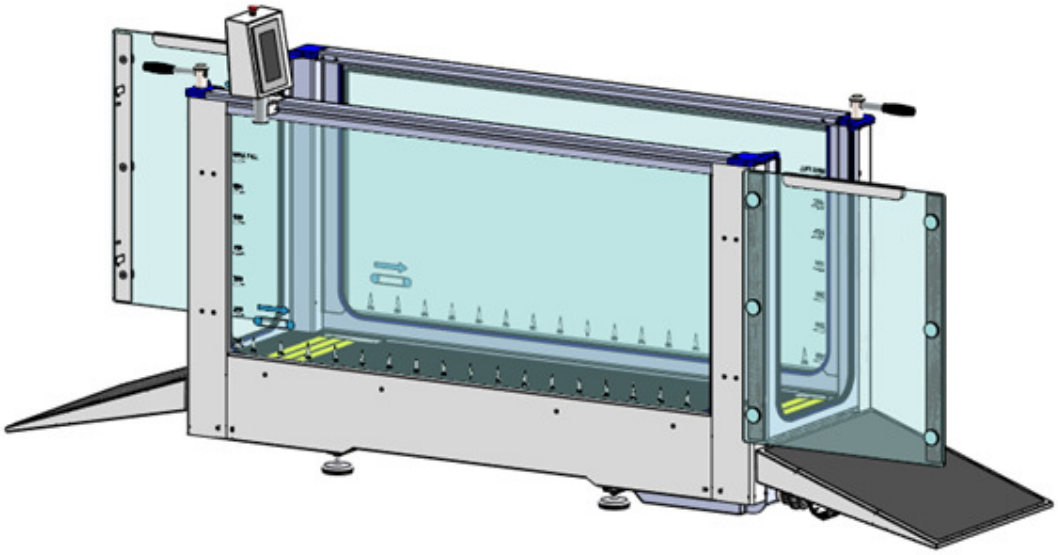


Figure 1-2

1.3 The control system

1.3.1 Operator control panel

The operator panel, mounted to the treadmill chamber, provides intuitive control of Water depth, Treadmill speed, Treadmill direction, Treadmill incline and Time/distance via a touch screen HMI.

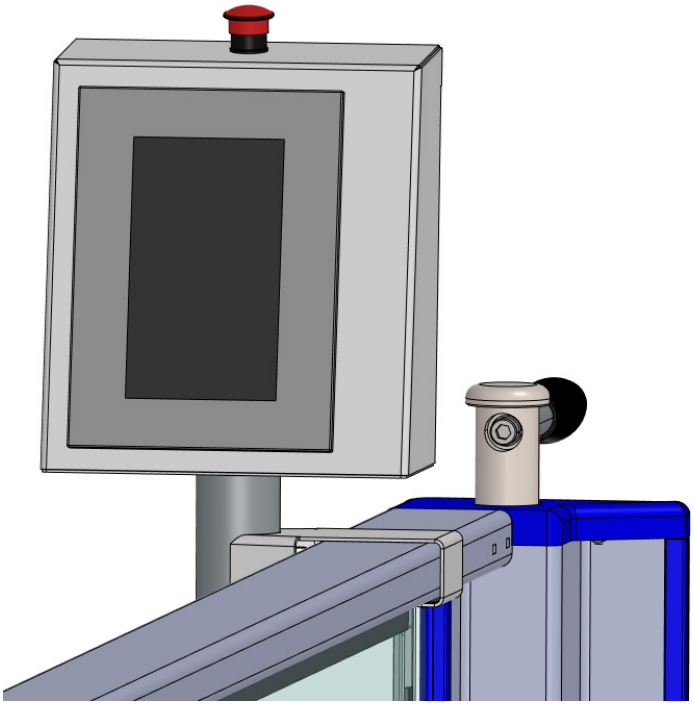


Figure 1-3. Touch screen HMI

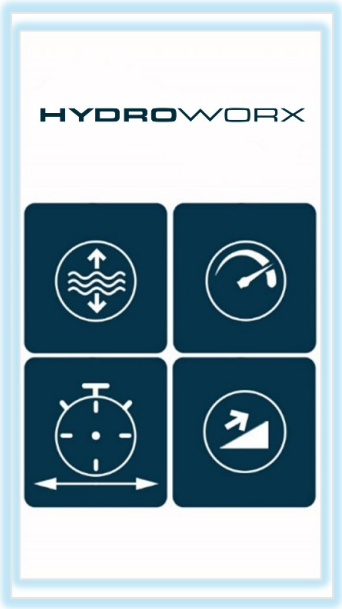


Figure 1-4. Example screen

1.4 Emergency stop

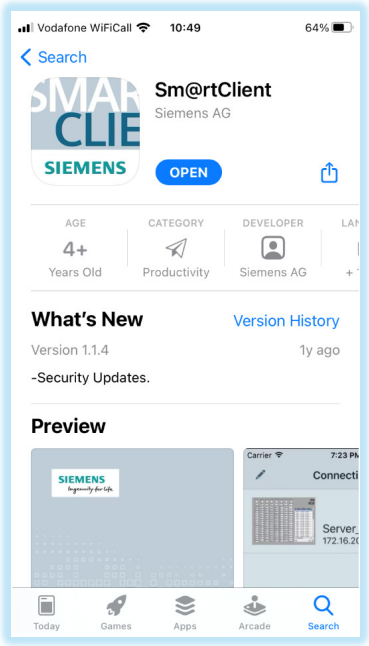
Pressing the emergency stop mounted on the operator panel and the main control panel stops the treadmill and evacuates the water from the chamber.

When pressed the emergency stop push button will latch in place.

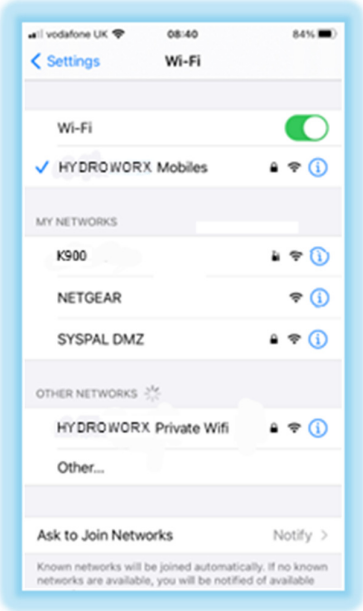
To reset the button, twist it in the direction indicated by the arrows on the button, this will allow it to pop out again. Once released press the '**Reset'** button on the operator screen.

1.4.1 Hand held mobile device

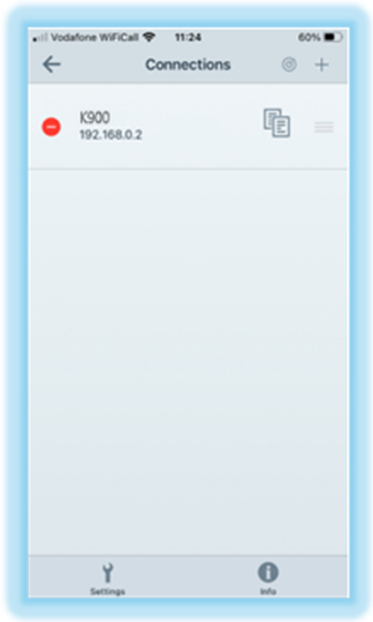
The K900 can be operated via an app on a mobile device.



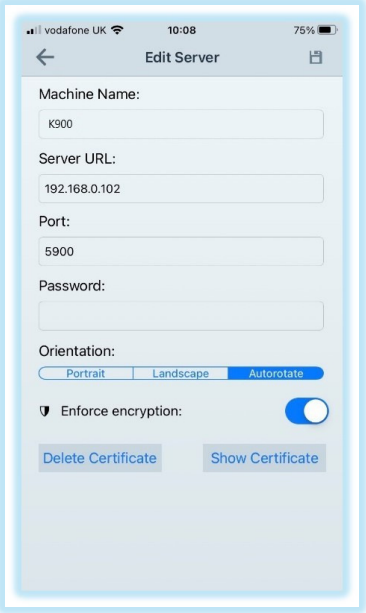
Download 'Siemens Sm@rtClient' app from App Store/Google Play.
(Do not select the Light version)



In the device Settings - Connect mobile device to K900 wifi network.



Open 'Siemens Sm@rtClient' app.
Select '+' icon



Add data as above
Enter Password **= Master**
Press Save.
Return to Connections.
Press K900 connection.
The operator screen will mirror on the device.

1.5 Water tank

The water storage tank is manufactured of high-density polyethylene. This means that there are no issues with corrosion or chemical resistance.

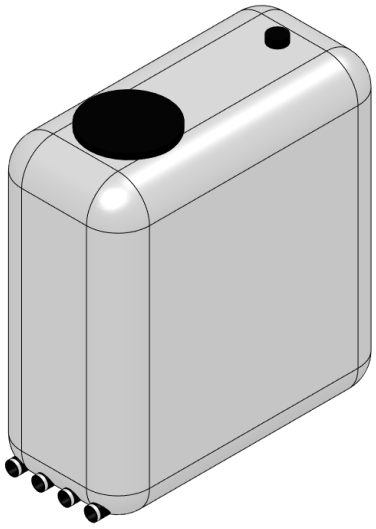


Figure 1-5. Water storage tank

1.6 Plant pack

The plant pack contains the pumps for filling and emptying the treadmill chamber, the circulation pump, the heater, the UV Clarifier, the sand filter and the main control panel.

The water temperature can be adjusted, using the heater control, to suit training or rehabilitation needs.

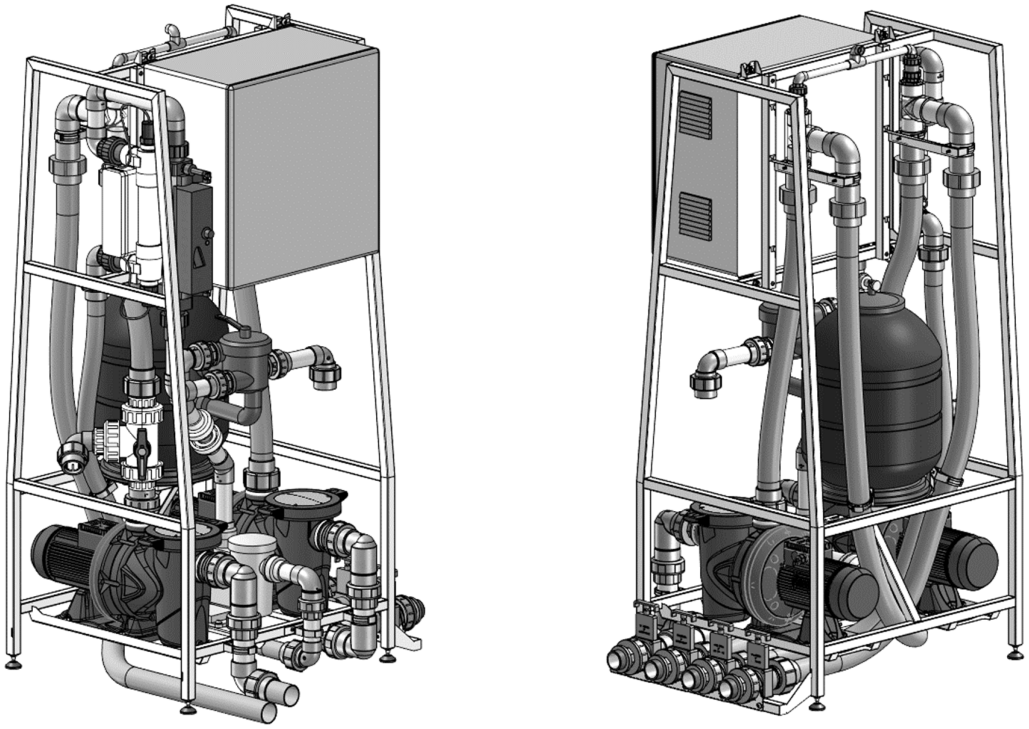


Figure 1-6. Plant pack

1.7 The water tank and plant pack

The water storage tank and plant pack combination allow fast filling and emptying of the treadmill chamber.

Water is constantly circulated, heated and filtered while the system is in use or standing idle. The system therefore has minimum water usage.

Water quality/chemistry is monitored and adjusted manually. Water chemical treatment is complemented with the UV Sanitizer.

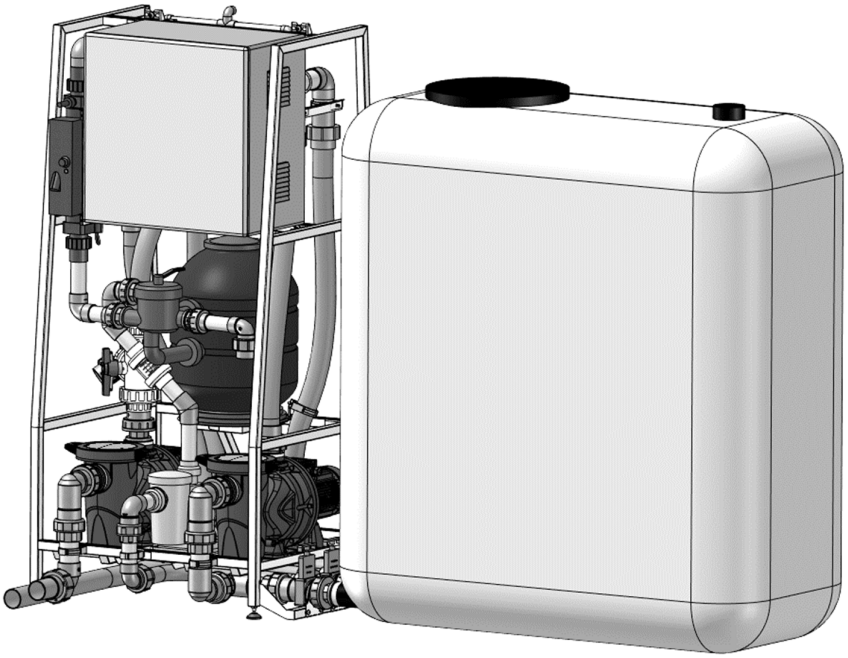


Figure 1-7

1.8 Water tank, plant pack and chamber

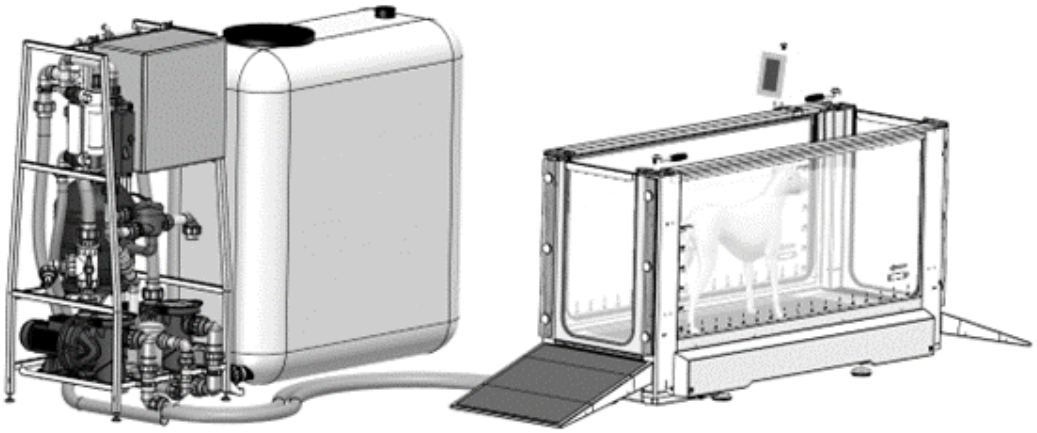


Figure 1-8

1.9 Access ramps

The access ramps are covered in a slip resistant covering with attractive hardwearing surface.

2. K900 USERS & SAFETY ⚠

The users of the K900 fall into four groups;

- Operators monitoring performance or administering treatment from outside or inside the chamber.
- Supervisors responsible for managing the operators and maintenance personnel.
- Individuals responsible for inspecting and maintaining the system.
- Individuals transporting, installing, commissioning, decommissioning or disposing of the system.

Each section of the instruction manual is addressed to specific groups of users. The table below shows which sections of the manual should be read and understood by each user group.

2.1 Responsibilities

Manual section	Operator	Supervisor	Maintenance personnel	Installation personnel
Introduction	✓	✓	✓	✓
K900 users & safety	✓	✓	✓	✓
Operation & control system	✓	✓	✓	✓
Cleaning	✓	✓	✓	
Routine Inspection & Maintenance		✓	✓	✓
Installation & commissioning				✓
Decommissioning & disposal				✓

The instructions for the K900 system are split into four sections:

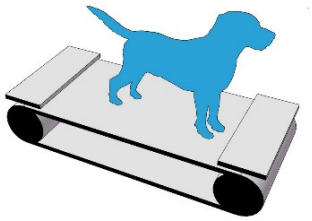
- K900 Users & Safety ⚠ - simplified safety instructions for day-to-day operation of the K900.
- Operation & Control System - for day-to-day operation of the K900.
- Routine Inspection & Maintenance - the technical details of the system, together with maintenance schedules.
- Installation & Commissioning - specialised instructions for installation and decommissioning.

2.2 Operator

Individual Operators must be assessed to ensure that they have a level of competence appropriate to their responsibilities.

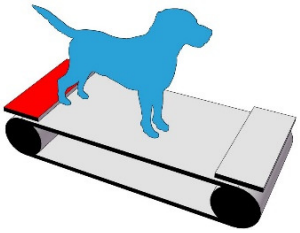
All operators must ensure that:

- they have read and understood the operating instructions before using the K900.
- they operate the K900 safely, in accordance with instruction and training given.
- all water spillages are cleaned up to avoid the possibility of a fall due to slipping.
- the K900 is operated at all times with full consideration to the animal's welfare.
- animals are supervised at all times.
- animals are on a lead and correctly positioned centrally on the treadmill walking belt.



- the animal is kept away from the treadmill end guards whilst the walking belt is moving, there is a danger of injury caused by the trapping point between guard and walking belt.





- if the animal approaches the end guard whilst the walking belt is moving, they press the emergency stop immediately.
- in case of a fault, stop the K900 and notify their supervisor of any problem.
- they ensure no mains voltage equipment, such as lighting, heaters or power tools are brought into the area.
- they understand how to stop the treadmill at any time by pressing the emergency stop button.
- they employ a safe system of work when around the treadmill chamber.

NOTE: Under no circumstances should an operator attempt to repair, adjust, or correct a fault with the K900.

2.3 Supervisors

Must have read and understood the operating instructions for the K900.

All supervisors must ensure that:

- all operators have completed a recognised canine hydrotherapy training course.
- they set and check the operating water temperature on the heater control panel.
- chemical levels in the water are correct after each treatment.
- all guards are present and fixed securely.
- the treadmill chamber is cleaned out at recommended intervals.
- the pump filter baskets on the plant pack are emptied at recommended intervals.
- they perform common troubleshooting procedures.
- they understand how to stop the treadmill at any time by pressing the emergency stop button.

2.3.1.1 Personal protective equipment (PPE) for the operator/supervisor

No personal protective equipment is required for operators/supervisors working outside the chamber.

2.4 Emergency stop function

There are two emergency stop buttons on the system. One is located on the treadmill chamber operator console. The second is located on the main control panel on the plant pack.

Pressing any of the emergency stop buttons stops all functions and drains the treadmill chamber.

When pressed the emergency stop button will remain depressed.



Before resetting the system following an emergency stop, ensure users are safe and the equipment is in a safe condition to restart.

To reset the button twist in the direction indicated by the arrows on the button, this will allow it to pop out again. Once released press the '**Reset'** button on the operator screen.

2.5 Water temperature

Water temperature should be selected according to the intended use. Guideline settings are 89.6 °F for rehabilitation, 80.6 °F for general exercise at walking speeds, and 77 °F for higher intensity training. The maximum water temperature is 104 °F. The risks to an animal from the water being too hot or cold must be evaluated by the person administering the treatment or exercise programme.

The water temperature must be checked at the beginning and end of every session.

2.6 Protection for the glass panels

In order to protect the glass panels from damage, users **MUST NOT**:

- climb over the sides of the chamber or push the glass panels inwards or outwards
- bring hard, sharp or metallic objects in or around the chamber
- wear jewellery or watches when using the chamber.

2.7 Protection from electric shock

In order to eliminate the risk of electric shock users **MUST NOT** bring mains voltage equipment, such as lighting or heaters into rooms where the treadmill chamber and plant pack are located.

2.8 Environment

To avoid fatigue and discomfort to operators working with the K900 for long periods, it is recommended that the environment around the system should be kept at 77 to 82.4°F and a maximum of 60% humidity.

2.9 Chamber drain time

The treadmill chamber takes approximately 2 minutes to drain from 24" water level. If the system is used for treating a medical condition, then the person administering the treatment must take this time into consideration when assessing the risks to the animal using the unit.

It is recommended that there is a period of time between users, approximately 10 minutes to allow the water to refresh, this period is dependent on usage.

2.10 Maintenance personnel

Maintenance must only be carried out by experienced personnel who are familiar with these instructions and with the operation of the K900. Maintenance personnel are responsible for implementing the routine inspection and maintenance schedule laid out in section 5.

All maintenance personnel must:

- exercise caution when using mains voltage equipment, such as lighting or power tools in the area.
- not use mains voltage equipment around the K900 unless the water is drained.
- ensure that the power supply to the main control panel is isolated before carrying out work inside it.
- employ a safe system of work when working around the K900.
- be familiar with the water chemical safety requirements.
- note additional safety requirements after water is added to the system.

2.10.1 Personal protective equipment (PPE) for maintenance personnel

No personal protective equipment is required for operators/supervisors working outside the chamber.

2.11 Installation personnel

Instructions for installation and decommissioning of the K900 must be made available to any contractors involved in the installation of the system, or its decommissioning and disposal at the end of its life.

2.11.1 Personal protective equipment (PPE) for installation personnel

Installation personnel are to wear PPE as required by national or regional regulations. Examples of PPE that may be required are:

- high visibility work wear
- safety shoes with toe protection
- face mask
- safety harness if working at a height
- gloves
- safety glasses
- ear defenders
- safety helmet.

In addition, when working with the chemicals used with the K900, all installation personnel must be familiar with the chemical safety requirements in Section 5.6.

3. OPERATION & CONTROL SYSTEM

3.1 Starting the HydroWorx

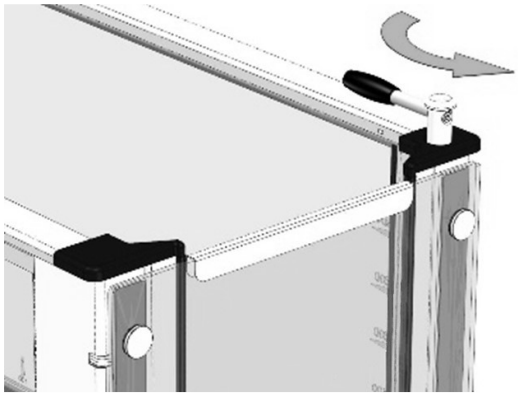
Normally the K900 power is left on, ensuring that the water is continually filtered and heated much like a swimming pool. If the power is switched off the procedure for switching on is described in section 3.3, Switching On/Off .

3.2 Door latch operation

The door is latched shut by the handle mounted on the door frame. The handle is rotated to clamp the door shut and is pressed down to release the door. This section shows the door operating procedure.

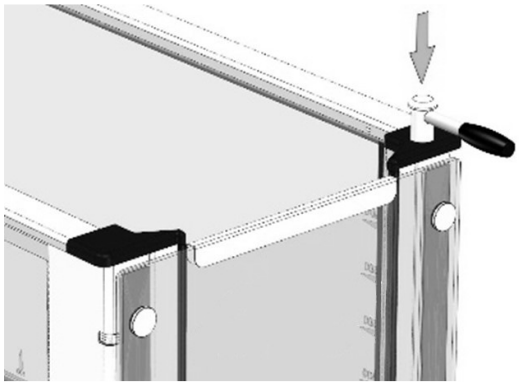
Ensure all the water has been evacuated from the chamber **before the door is opened**. See the **water depth menu** screen in Section 3.5.

Opening the door



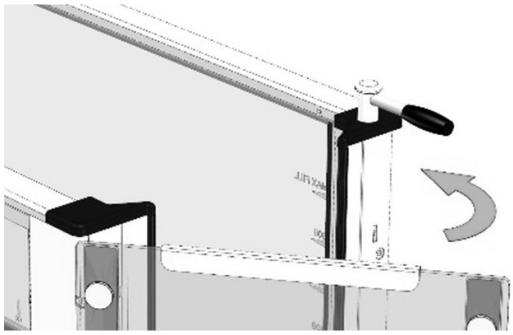
With the door handle in the **'LOCKED'** position, rotate the handle anti-clockwise 180 degrees.

The door is now **'UNLOCKED'** on the safety latch.



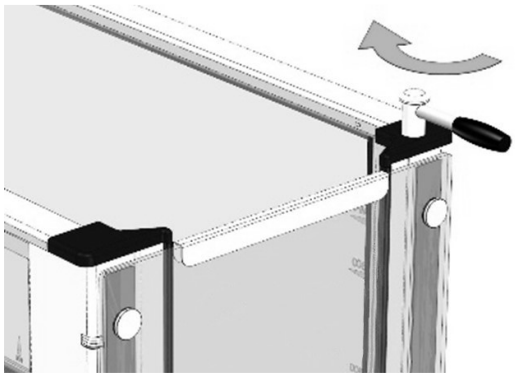
Push down on the handle to release the safety latch.
Swing door outwards to open.

Closing the door



With the door handle in the **'UNLOCKED'** position, swing door firmly closed.

The handle will pop-down and the door will be retained on the safety latch.



Rotate the handle clockwise 180 degrees. (Some resistance may be felt).

The door is now **'LOCKED'.**

NOTE: If the door handle is rotated whilst the treadmill is full of water, the safety latch will hold the door closed, water will however leak freely around the door seal. If this happens, quickly turn the handle to re-seal the door. DO NOT press the handle down to release the safety latch. The door will open rapidly, allowing the water in the chamber to flood out. This could cause the occupant of the chamber to be swept out with the water, possibly causing them serious injury.

It is recommended that the doors are left in the locked position when the treadmill is not in use.

3.3 Switching On/Off

Before switching on, a number of checks must be made.

3.3.1 Pre-switch-on checks



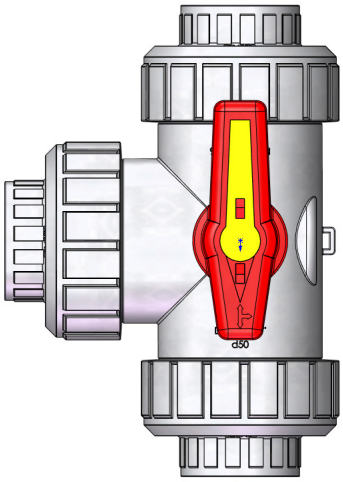
Ensure the lever on top of the sand filter multiport valve is set to 'FILTER' as shown.



Check that the blue switch located on the front of the main control panel, is set to **FILTER.**



Check that all the gate shut-off valves on the water storage tank are OPEN as shown.



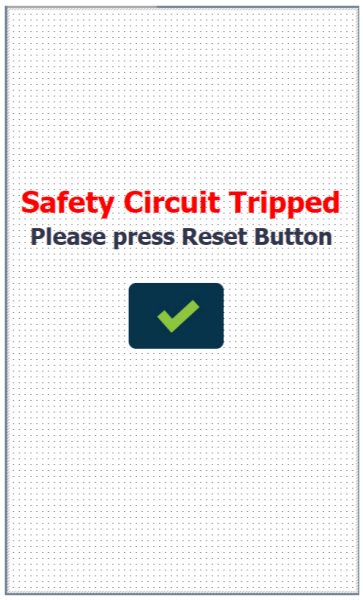
Check red lever on the manual diverter valve is set to closed to waste pipe as shown.

3.3.2 Switching on



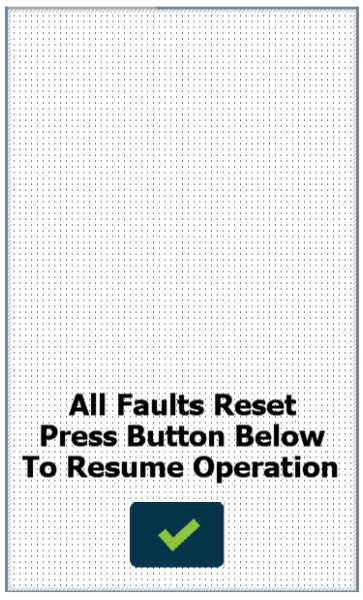
Once the Pre-switch on checks have been carried out, the K900 can be switched on.

Rotate the main power switch on the main control panel, clockwise to the **'ON'** position.



The screen on the operator panel will display an information message as shown.

Press the **RESET** button.



The screen on the operator panel will display an information message as shown.

Press the **RESET** button.

The system is now ready for use.

3.4 Water temperature

Water temperature should be selected according to the intended use. Guideline settings are 89.6 °F for rehabilitation, 80.6 °F for general exercise at walking speeds, and 77 °F for higher intensity training. The maximum water temperature is 104 °F. The risks to an animal from the water being too hot or cold must be evaluated by the person administering the treatment or exercise program.

The water temperature must be checked at the beginning and end of every session.

3.4.1 Setting water temperature

The desired water temperature is set from the heater temperature set point dial.

Time must be allowed for the adjustment of the water temperature, approx 34.7°F per hour. Also note that the temperature will not adjust below the ambient temperature.

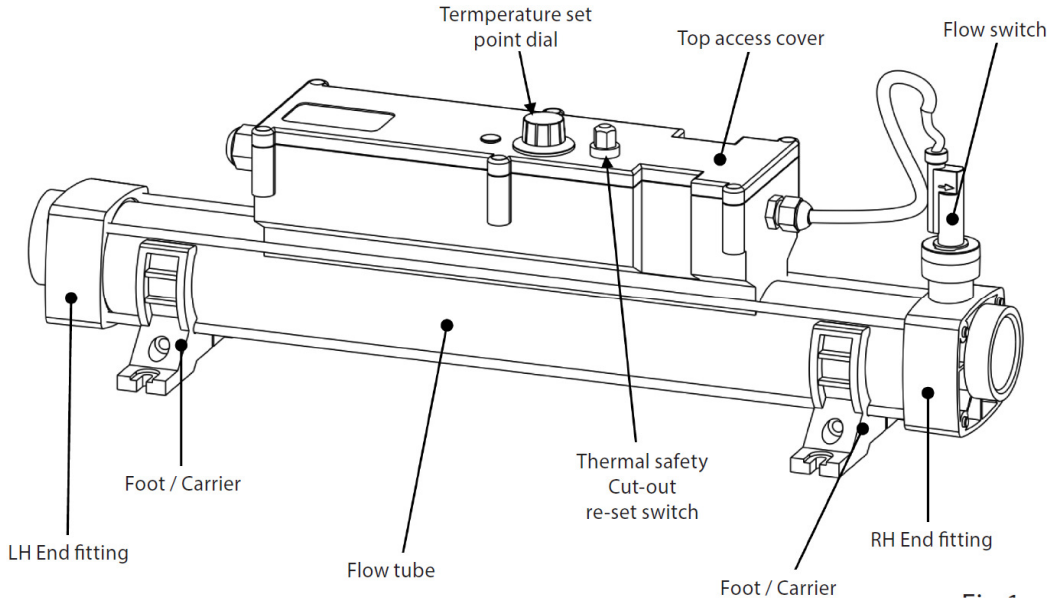
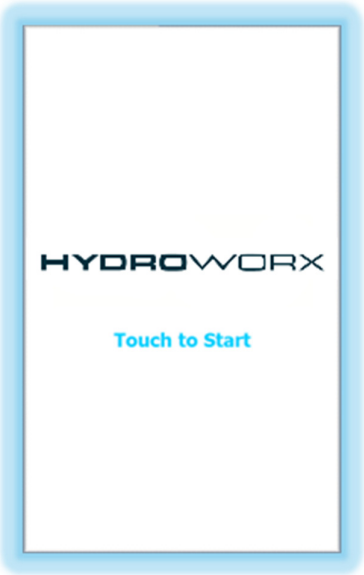


Figure 3-1

3.5 Touch screen operation

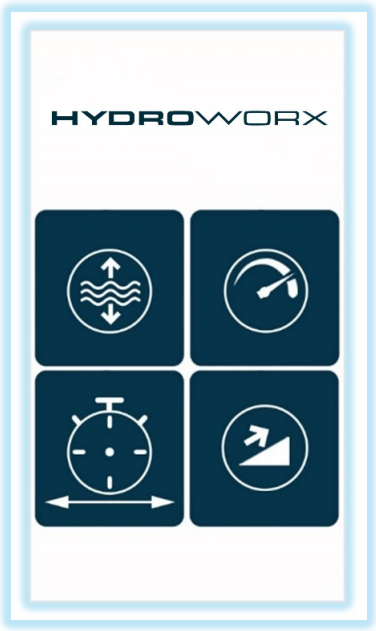
3.5.1 System operation

1. The Standby screen



Touching the HydroWorx logo will display welcome screen.

2. The Welcome screen



Four options are displayed, touching any of the windows will display the function.

 Water depth

 Treadmill speed

 Session time/distance

 Treadmill Incline

3. Warning screens



A reminder will display.
Has the water quality been checked today?

Check water quality is within allowable limits, see Section 5.

Confirm this selection by pressing 



A reminder will display.
Are both doors closed?

Confirm this selection by pressing 

3.6 General screen features

Pressing any of the function icons at the top of the screen will display the relevant function at any time.



Pressing any of the buttons at the bottom of the screen at any time will:



End session, Stop treadmill and empty chamber**.** An End Session confirmation screen will show.



Pause session.

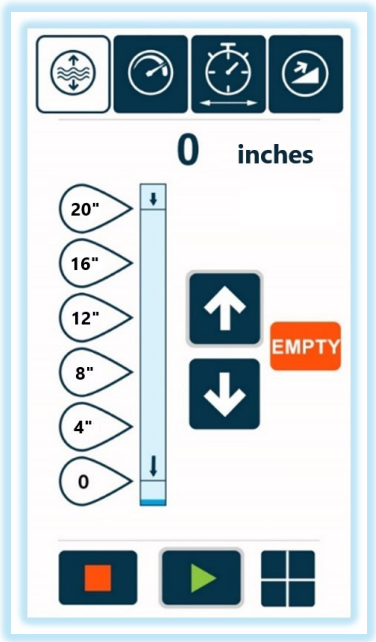


Re-start session.



Return to OVERVIEW screen.

1. Water screen




Water depth.

Before filling, measure the animal to establish the suitable water height for treatment.

To fill to the desired pre-set water depth, press the appropriate water symbol on the left-hand side of the screen.

The water fill can be stopped at any time by pressing


or


The water depth can be adjusted up or down by pressing


or


The actual water depth is displayed at the top of the screen.

2. Treadmill screen – Training speeds



 Treadmill speed.

Before selecting a treadmill speed:

 Ensure the animal will be walking forward.

 Ensure the animal is clear of the treadmill end guards.

To select the desired treadmill speed, press the appropriate pre-set speed button on the left-hand side of the screen.

To start the treadmill press 

The treadmill can be stopped at any time by pressing  or 

The treadmill speed can be changed by selecting an alternative pre-set speed or adjusted up or down by pressing



The actual treadmill speed is displayed at the top of the screen.

Optional therapy pre-set speeds can be selected by pressing 

3. Treadmill screen – Therapy speeds.



 Treadmill speed.

Before selecting a treadmill Therapy speed:

 Ensure the animal will be walking forward.

 Ensure the animal is clear of the treadmill end guards.

To select the desired treadmill speed, press the appropriate pre-set speed button on the left-hand side of the screen.

To start the treadmill press 

The treadmill can be stopped at any time by pressing  or 

The treadmill speed can be can be changed by selecting an alternative pre-set speed or adjusted up or down by pressing



The actual treadmill speed is displayed at the top of the screen.

Optional training pre-set speeds can be selected by pressing 

4. Treadmill screen – Reverse speeds



Treadmill speed.

From the treadmill speed screen, walking belt reverse direction can be selected by pressing . The belt must be stationary for this selection to function.

Confirm the selection by pressing

Ensure the animal will be walking forward.

Ensure the animal is clear of the treadmill end guards.

The reverse screen operates as the forward screens.

To return to forward direction press . The belt must be stationary for this selection to function.

5. Timing screen



Treadmill run time.

The treadmill timing screen displays the time elapsed and distance travelled for the current session.

The time and distance can be reset by pressing RESET.

The TIME can be set to a **COUNTDOWN** mode.

Pressing the minutes and seconds figures individualy on screen will enable the countdown time to be set via a pop-up key pad.

Start treadmill as normal, the time will count down.

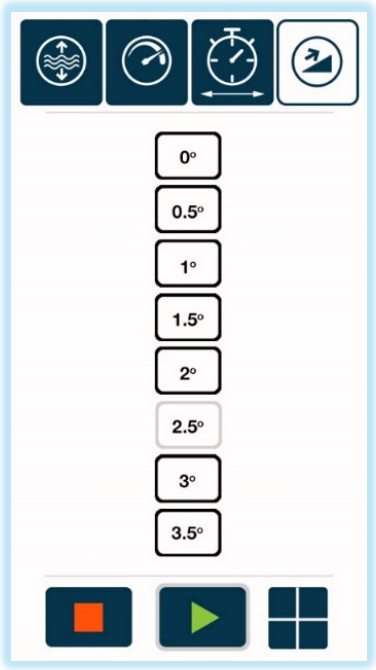
If the session is paused at any time the count down will hault and continue once the session is re-started.

The countdown time can be revised at any time.

The END SESSION prompt will display at the end of the countdown.

To exit the COUNTDOWN mode, END SESSION must be selected.

6. Incline screen



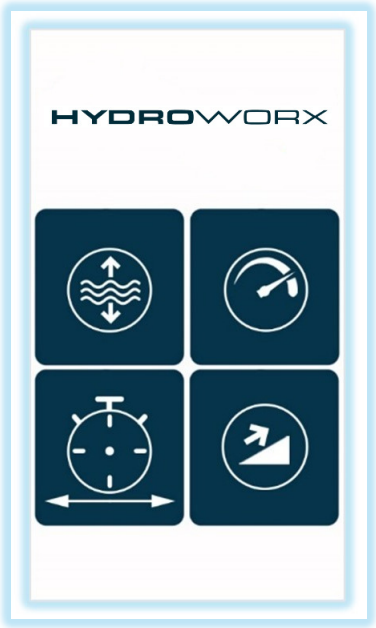
 Treadmill incline.

From the treadmill incline screen, the walking belt gradient can be selected by pressing the pre-set degrees buttons.

 Ensure the animal is clear of treadmill end guards before operating the incline.

The pre-set buttons have a press and hold function.

7. Overview screen



 Menu overview.

Pressing the overview button will display all the session features.

- Water depth.
- Treadmill speed.
- Session time and distance.
- Incline gradient.

Touching any of the windows will display the function.

3.7 Touch screen settings and administration

From the Introductory menu, the settings and administration and maintenance screens can be accessed by pressing The HydroWorx logo

3.7.1 General screen features

Pressing any of the function icons at the top of the screen will display the relevant function.



Pressing  button at the bottom of the screen at any time screen will return to operation screens.

1. Units screen.



 Unit selection.

From this screen the user can select the units displayed on the operation screens.

Unit options are for water depth and treadmill speed. The selected units are indicated in light blue.

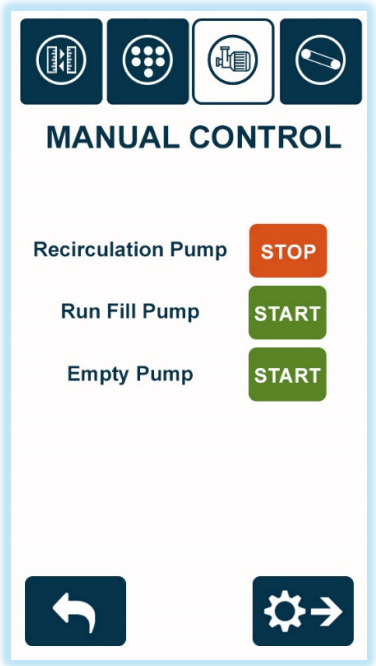
2. Setting IO screen.



 IO Status

For maintenance purposes, from this screen, the electrical Inputs/outputs can be viewed.

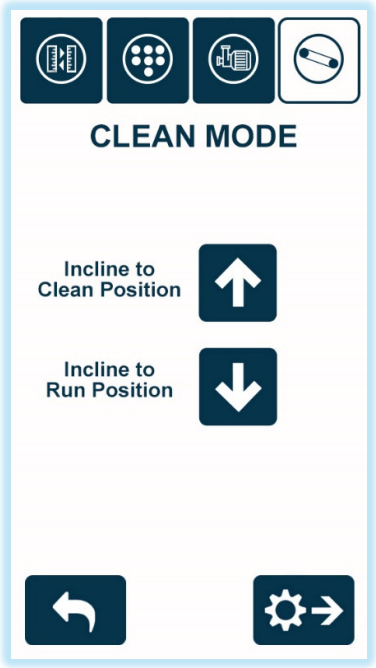
3. Manual control screen.



 Manual control

Ror maintenance purposes, from this screen, the water pumps can be operated individually.

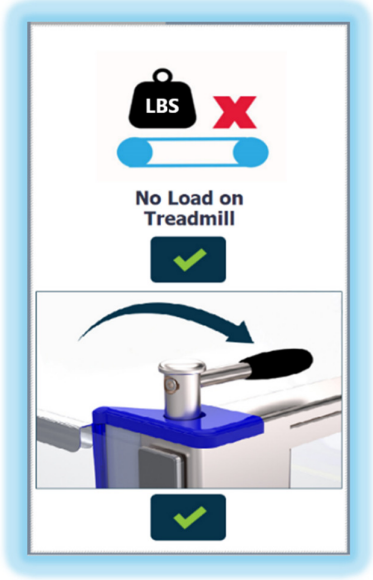
4. Clean mode screen.



Clean mode.

For cleaning purposes, the treadmill can be powered up into a raised position to allow access to the underside.

Press and holding the button will raise the treadmill. Treadmill will stop in clean position.

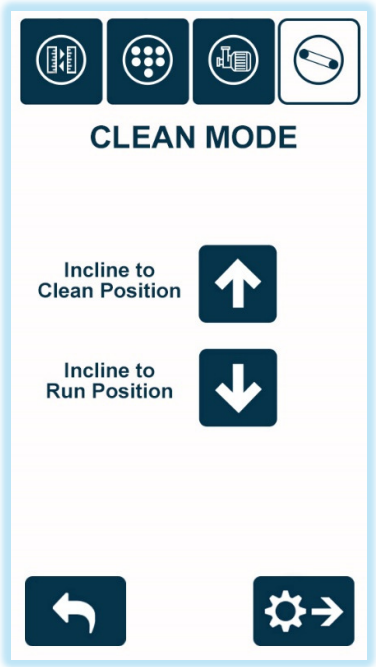


Before using this function, it is important to ensure there is no load on the treadmill.

Confirm this by pressing

Before using this function, it is important to ensure doors are closed.

Confirm this by pressing



Press and holding the button will lower the treadmill. Treadmill will stop at level positon.



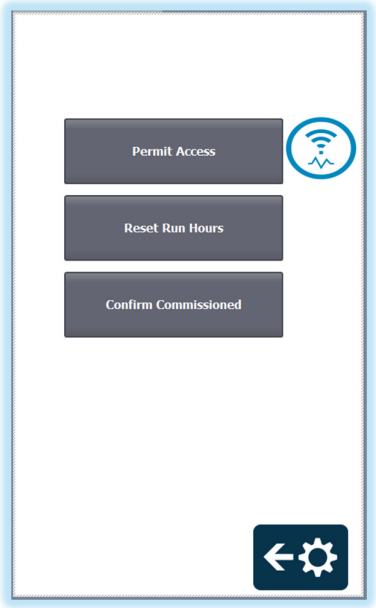
Before lowering treadmill ensure area below is clear of any cleaning equipment.

Confirm this by pressing

Before using this function, it is important to ensure doors are closed

Confirm this by pressing

5. Maintenance



Pressing  button at the bottom of any of the screens will show the maintenance screen.
These functions must only be used when in communication with a HydroWorx service engineer.

The K900 has a remote diagnostics feature, If this symbol is displayed on any screen, remote access has been permitted.



4. CLEANING

4.1 Personal protective equipment (PPE) to be used when cleaning

Rubber gloves and goggles are recommended when using the cleaning materials listed below.

4.2 Materials used in construction

Component	Material
All fabricated parts, framework & guard panels Machined components	Stainless steel, grade 316 (1.4401) Stainless steel, grade 304 (1.4301)
Components	Glass, Rubber, Plastic

NOTE: Although electrical components have a high IP rating, it is recommended that they are not directly hosed with water jets. All other external components are suitable for wash down with low pressure water jets.

4.3 Cleaning materials

The following cleaning materials should be used for general cleaning of the chamber, the storage tank and surrounding area, as they are compatible with the water treatment chemicals.

- A cloth dampened with a water solution of free chlorine or bromine, with a concentration of 18-37 mg/Gal.
- A rubber squeegee should be used to remove drips from the chamber glass after cleaning.
- A Wet & Dry vacuum cleaner

If brushes or mops are used, care must be taken not to knock the glass panels with the handles when cleaning.

Should a more abrasive cleaner be required for particular areas, such as the chamber waterline, sodium carbonate or bicarbonate can be placed on a damp cloth and applied sparingly. Always test a small area to ensure compatibility.

Some cleaning materials (especially detergent based cleaners) may react with the water treatment chemicals or be incompatible with the K900 materials. These must not be used unless they can be shown to be compatible.

NOTE: Abrasive wipes should not be used on any surfaces.

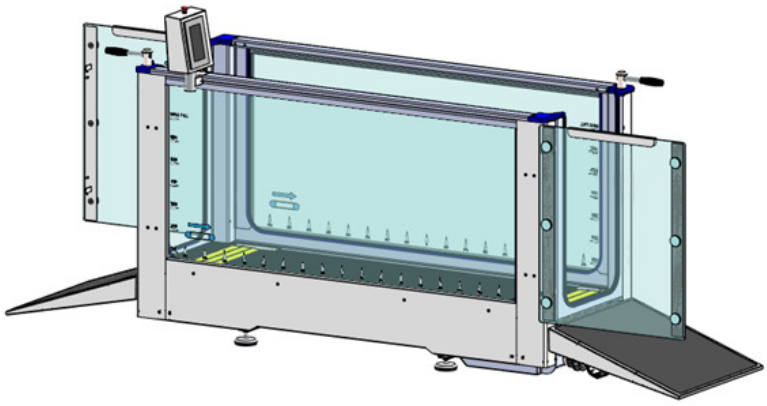


Figure 4-1. Chamber

4.4 Daily cleaning procedure

The inside and outside of the chamber must be cleaned daily with fresh water, including:

- glass side panels and door
- stainless steel surfaces
- the floor areas surrounding the treadmill
- touch screen control
- the water pump strainer baskets should be inspected, emptied and cleaned if necessary, refer to section 4.9, Cleaning the pump filter baskets.
- The inside of the chamber under the treadmill.

4.5 Weekly cleaning procedure

The inside surfaces of the storage tank including the underside of the lid should be manually cleaned once a week. This cleaning should coincide with the weekly draining of the system.

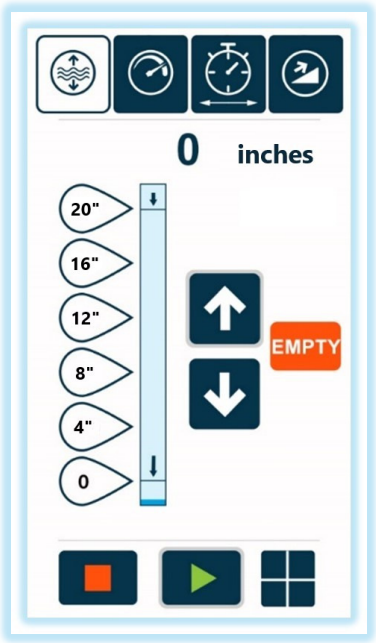
4.6 Cleaning chamber glass and doors

It is good practice to wipe down the inside of the glass with a cloth and clean water after each use. It is recommended that the treadmill chamber is fully cleaned following each day's use.

4.7 Cleaning underneath the treadmill

It is recommended that a wet/dry vacuum cleaner is used to clean underneath the treadmill to remove any debris.

4.7.1 To fully empty treadmill chamber of water.



Go to the water screen.

Press the EMPTY button to empty the chamber of water.

4.7.2 Move treadmill in to clean mode position

See section 3.7,Touch screen settings and administration for cleaning mode instructions.

NOTE: No attempt should be made to manually lift the treadmill beyond its cleaning position. This could result in personal injury and damage to the equipment.

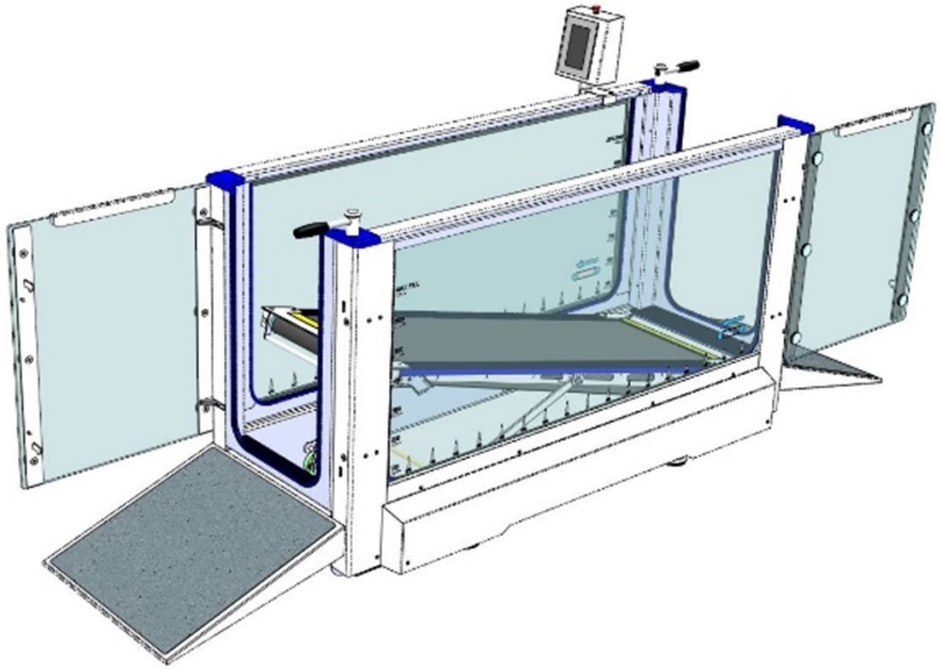
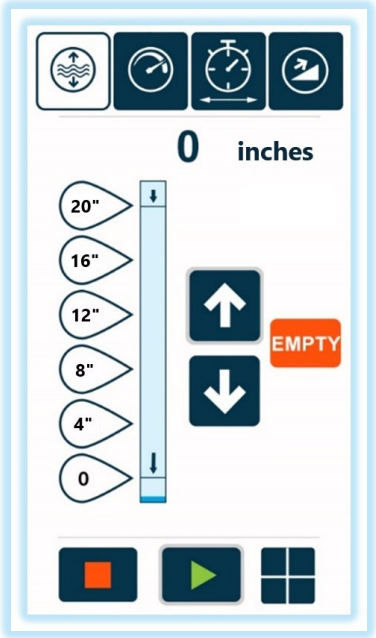
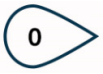


Figure 4-2. Treadmill in cleaning position

4.7.3 Refilling treadmill chamber following cleaning.



IMPORTANT – Ensure both doors are closed and locked. See section 3.2, Door latch operation.

To fill to the pre-set water depth, press the 

water symbol on the left-hand side of the screen.

The water fill can be stopped at any time by pressing



4.8 Cleaning the sand filter

The sand filter is situated on the plant pack. Periodically the sand inside will need to be cleaned by following the **Backwash** & **Rinse** procedure. It is recommended this procedure is followed once a week minimum or following every 40 dogs, whichever occurs first.

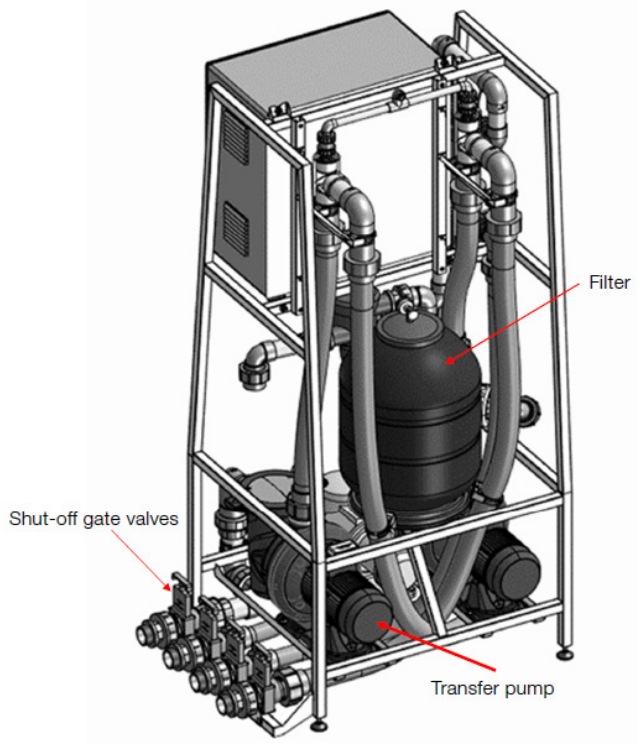


Figure 4-3

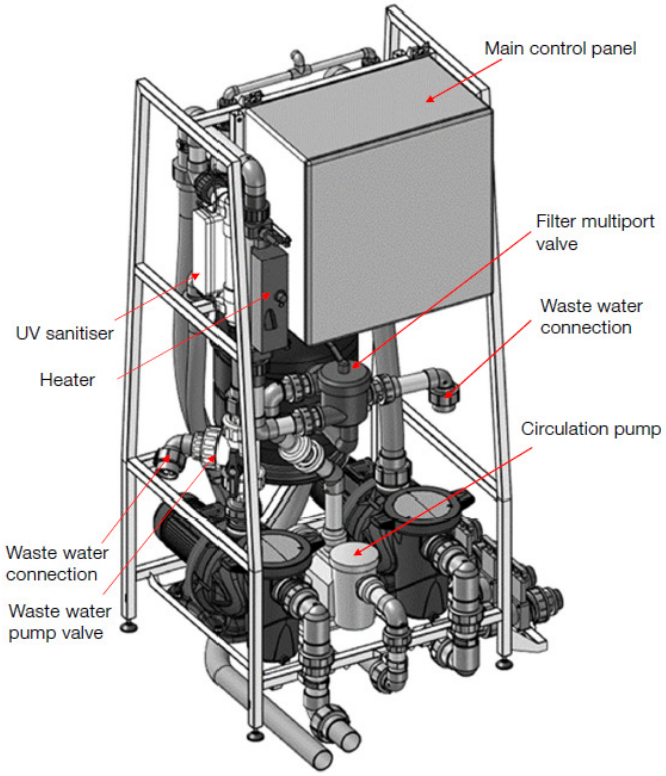


Figure 4-4

4.8.1 Backwash and rinse procedure



Ensure the system is not in use.

On the main control panel, move the **blue** **switch** to **Clean**. The recirculation pump will stop. This can be checked via the pressure gauge on the sand filter.



On the sand filter, move the **multiport valve** to **BACKWASH**.



Press the **Backwash** button.

Wait for 40 seconds for the **pump** to stop.



Move the filter multiport valve to RINSE.



Press the **Rinse** button.

Wait for 10 seconds for the **pump** to stop.



IMPORTANT – Return the filter multiport valve to FILTER.



Move the **blue** **switch** back to f**ilter**.

Top up the Water storage tank with fresh water.

The system is now ready to use.

4.9 Cleaning the pump filter baskets

Each of the three pumps has a built-in filter in the form of a plastic basket. This can be found under the clear lid on top of the pump.

To prevent excessive water loss, it is recommended that these filters are cleaned when the water tank and treadmill chamber are empty. Filters may need to be cleaned more frequently depending on hair loss of the animal. As long as the pumps are not running, a minimal amount of water is lost when removing the baskets.







1. Ensure the main switch on the control panel is switched off.
2. Unscrew the clear lid on the top of the pump.
3. Remove the filter basket and empty any debris.
4. Change the filter sock fitted to the wire cage in the empty pump basket as required.
5. Replace the basket and re-fit the clear lid to the top of the pump.

4.10 Changing the water in the water storage tank.



Ensure the system is not in use, and all the water is in the storage tank.

Ensure the waste outlet from the filter on the plant pack is directed towards the waste water drain.

On the main control panel, move the **blue switch** to **CLEAN**.
The recirculation pump will stop, this can be checked via the pressure gauge on the filter.



On the sand filter, move the Multiport Valve lever to the waste position.



On the main control panel, move the blue switch to FILTER.
The recirculation pump will start, this can be checked via the pressure gauge on the filter.

The filter valve will divert the water to the waste water drain.

Continue until the water storage tank is empty.



Once the water tank is empty move the switch back to CLEAN.
The recirculation pump will stop.



IMPORTANT: Return the filter multiport valve to FILTER.

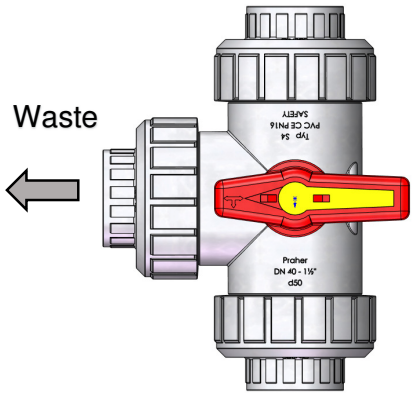
Refill the Water storage tank with fresh water and return the blue switch to FILTER.

The system is now ready to use.

If all the water in the system is changed refer to section 5.6.7, Pool test kit for water chemicals.

4.11 Dumping contaminated water from treadmill chamber

Ensure the waste water outlet from the Plant Pack is directed towards the waste water drain.

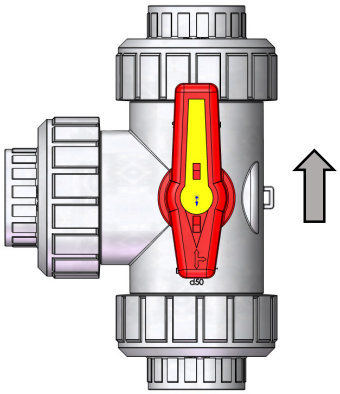


Set the red level of the manual diverter valve open to waste pipe as shown.

Navigate to the water level screen on the operator touch screen. 

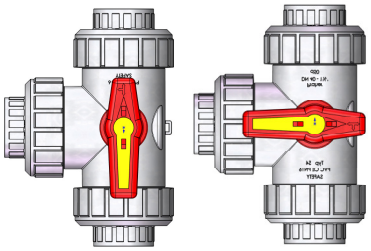
Press the 

chamber button, allow treadmill chamber to empty as normal.



Re-set red lever on the manual diverter valve to closed to waste pipe as shown. (normal position)

Do not set vave to these positions:



Check water level in water storage tank, add fresh water as required.

The System is now ready to use.

5. ROUTINE INSPECTION & MAINTENANCE

This section details the maintenance schedules for the component parts of the HydroWorx system.

5.1 Maintenance personnel - safety

Maintenance must only be carried out by experienced personnel who are familiar with these instructions and with the operation of the K900.

- Do not bring mains voltage equipment, such as lighting, heaters or power tools into the chamber or plant pack/storage tank areas.
- Do not use mains voltage equipment around the plant pack, storage tank and pipework unless the tank and pipework system have been drained.
- Ensure that the power supply to the main control panel is isolated before carrying out work inside it.
- Be especially careful not to knock the glass when using tools in or around the chamber.
- Drain the storage tank before working inside the water storage tank.
- Shut off the water supply to the water storage tank when working inside it.
- Employ a safe system of work when working inside the water storage tank.
- All maintenance personnel must be familiar with the chemical safety requirements.

5.2 Maintenance schedules (Water management)

Testing and recording of residual disinfectant and pH levels should be undertaken before the Hydro system is used each day. A logbook should be available for recording results. Records should be kept for a minimum of 5 years.

5.2.1 Daily inspection

Personnel must carry out a visual inspection every day to ensure that:

- all guarding is securely in position and in good condition
- both end ramps are securely in position
- the machine is running smoothly with no evidence of excessive noise or vibration
- the door seal is clear of dog hair and debris
- the pH and chemical levels in the water are correct
- the glass is wiped clean after use
- the water is at the required temperature.

5.2.2 Weekly inspection

Personnel must carry out a visual inspection every week to ensure that:

- all guarding is securely in position and in good condition
- the machine is running smoothly with no evidence of excessive noise or vibration
- water should be changed weekly
- the pump filters are cleaned of any debris
- lubricate pump 'O' rings
- the glass in the treadmill chamber is cleaned
- the sand filter has been backwashed and rinsed.

5.3 Maintenance procedures

This section summarises the maintenance schedules for the component parts of the K900 system as specified by the manufacturers. The manufacturers' instructions for each component are included in this manual. These must be referred to for more detail before carrying out any maintenance work.

5.3.1 Water system plant pack

The figure below shows the layout of the component parts comprising the standard plant pack. The hydraulic circuit diagram can be found at the end of this manual.

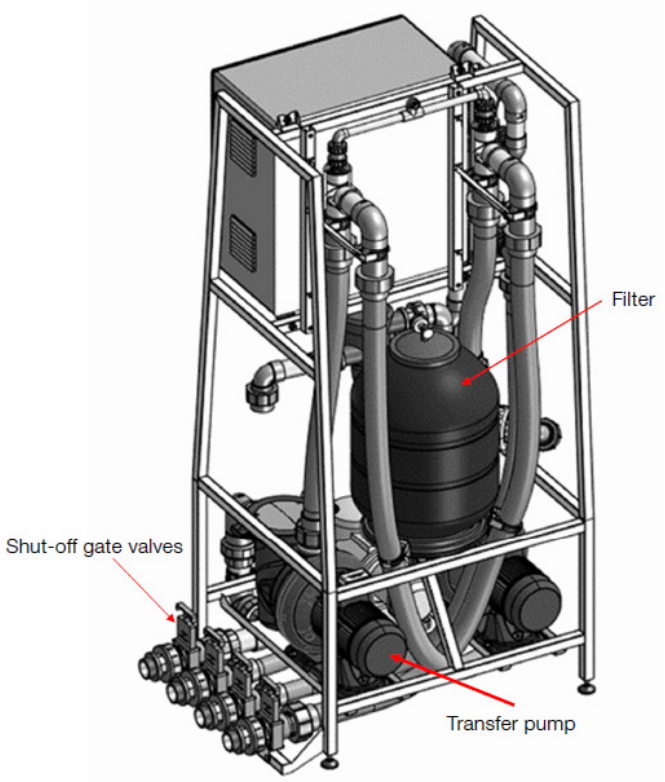


Figure 5-1

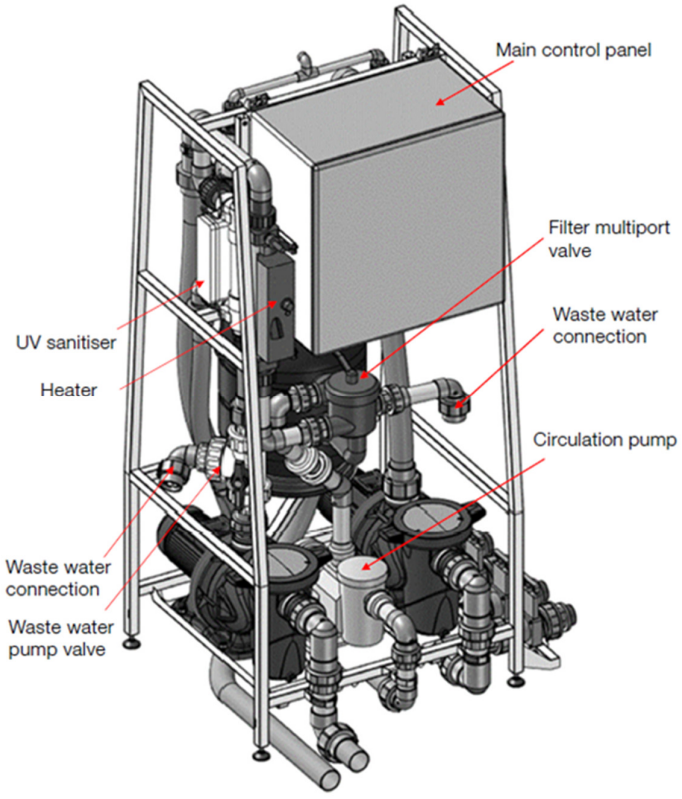


Figure 5-2

5.4 Routine inspection and maintenance of plant pack components

Component	Routine inspection	Routine maintenance	PPE required
Sand filter	None	Replace the filter media in the sand filter every five years. At this time the rubber seals should be replaced also. The sand in the filter is a zeolite type, Zeoclere 30. The procedure is as follows: 1. Drain the system. 2. Disconnect the sand filter from the pipework. 3. Drain the water from the filter using the drain valve at the bottom. 4. Remove the valve from the top of the filter. 5. Manually remove the sand. 6. Refill the filter to the halfway point with clean water. 7. Refill the filter to the halfway point with Zeoclere filter medium. 8. Backwash for 3 minutes 9. Rinse for 2 minutes 10. Fill the filter to the top with Zeoclere. 11. Backwash again for 3 minutes. 12. Rinse again for 2 minutes.	When handling dry Zeoclere a breathing mask and gloves are required, for protection against dust.
Heater	None	None	None
Transfer pumps	None	Clean the strainer basket and lubricate 'O' ring weekly. o-ring lube is a Teflon or Silicone base to keep the rubber soft and supple	None
Recirculation pump	None	Clean the strainer basket and lubricate 'O' ring weekly. o-ring lube is a Teflon or Silicone base to keep the rubber soft and supple	None
UV Sanitiser	None	None	None

5.4.1 The K900 exercise chamber

The figure below shows the exercise chamber layout, the routine inspection points are detailed in the table below.

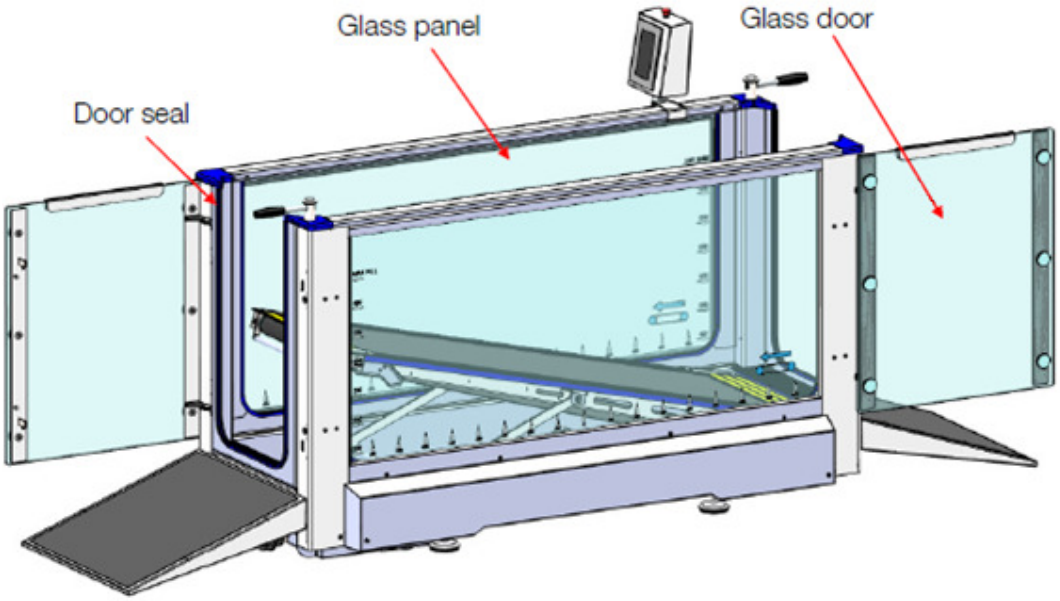


Figure 5-3

5.5 Routine inspection and maintenance of chamber components

Component	Routine inspection	Routine maintenance	PPE required
Door seal	Check seal daily to ensure it is free from damage and no leaks are occurring. Replace if damaged or if leaks around door are evident.	None	None
Glass panels & door	Check glass panels weekly for chips, cracks and deep scratches, particularly on the top and bottom edges of the glass. If present, stop using the K900 and contact the HydroWorx team for advice.	None	None
Treadmill	Check the surface of the treadmill belt monthly for signs of wear. If present, stop using the K900 and contact the HydroWorx team for advice.	None	None

5.6 The water treatment system

5.6.1 Water treatment chemicals - safety

The chemicals used in the water treatment are Bromine or Chlorine granules, pH Increaser and pH reducer. Safety data sheets for these chemicals are included in the appendices. The warning labels applicable to these chemicals are shown below:



Chlorine & bromine Granules



pH Reducer



pH Increaser

NOTE: The hazards from these chemicals can vary depending on their form and concentration, so it is important to read the specific material data sheet supplied with the particular brand of chemical being used using for safety information.

5.6.2 Personal protective equipment (PPE) for handling and mixing chemicals

When handling these chemicals, the following PPE must be worn:

- Chemical proof rubber gloves
- Splash proof goggles
- Dust mask when handling the chemicals in a granular or powder form
- An eye-wash facility, located near to where the chemicals are handled.

5.6.3 Handling safety

Wash hands after handling chemicals.

Never allow the chemicals to come into contact with skin.

Never eat, drink or smoke when handling chemicals.

Always make sure the relevant material data sheets are prominently displayed at the area where the chemicals are used.

5.6.4 Chemical storage safety

Never store pool chemicals, particularly chlorine, near any gasoline, kerosene, oil, brake/hydraulic fluid, grease, oil-based paint etc. Keep pool chemicals in a completely separate, secure place away from other stored items.

Always store pool chemicals in their original containers and in a cool, dark, dry place.

Always make sure the relevant material data sheets are prominently displayed at the storage area.

5.6.5 Chemical warning signs

Warning signs indicating what PPE is to be used must be displayed.

Chemical storage containers must be labelled with the name of the chemical and the appropriate hazard warning sign.

Emergency actions for a particular accident/incident must be clearly visible and easily understood.

5.6.6 Testing and maintaining chemical levels

The K900 is supplied with a pool test kit and a starter pack of chemicals. The K900 can be used with either Chlorine or Bromine depending on preference of user and local authority by laws.

Replacement chemicals are available from the HydroWorx distributor or a local swimming pool/spa shop.

5.6.7 Pool test kit

Using the pool test kit provided and following the instructions in this kit, test the water in the storage tank check for Total Alkalinity, pH & Bromine/Chlorine levels after each treatment. If the **Total Alkalinity** is found to be outside the ideal range **80 – 120ppm** it is recommended that the water in the HydroWorx is changed. Refer to Section 4.10, Changing the water.

If all the water in the system is changed add **3/8 ounce** of **BROMINE** or add **1/5 ounce** of **CHLORINE** to the fresh water.

5.6.8 Adjusting the pH level

Correct water pH ensures that the disinfection efficiency of chlorine or bromine is optimal.

The ideal range for pH between 7.2 and 7.6, adjustments need to be made manually.

To increase full storage tank pH by 0.1 add approximately 7 grams of pH Increaser.

To decrease full storage tank pH by 0.1 add approximately 7 grams of pH Reducer.

Correct water pH ensures that the disinfection efficiency of chlorine is optimal.

The safe range of pH should be between **7.2** and **7.6**.

- If the pH is above this then use **pH minus** powder.
- If the pH is below this then use **pH plus** powder.

To **increase** the pH use the chemical **pH plus**. This should be added at the rate of **1/4 ounce** per full storage tank for an increase of **0.1** on the pH scale.
The powder should be added to the storage reservoir and mixed thoroughly. The powder **should not** be seen after mixing.

To **decrease** the pH use the chemical **pH minus**. This should be added at the rate of **1/4 ounce** per full storage tank) for a decrease of **0.1** on the pH scale.
The powder should be added to the storage tank and mixed thoroughly. The powder **should not** be seen after mixing.

Once the pH has been adjusted, pump the water from the storage tank to the treadmill chamber and back again. This will help to mix the chemical and the water.

5.7 Bromine systems

5.7.1 Adjustment

a) Maintain Bromine levels between **2** and **6ppm**.

b) With fresh water add **3/8 ounce** per full storage tank. Wait 30 minutes and test. This should give a reading of about **5 ppm**. Bromine is normally available in **20g** tablet form, in this form add half a tablet.

c) To maintain Bromine level, add as required, but in very small quantities **3/16 ounce** or **¼ tablet** to increase by **1 ppm** as required.

5.7.2 Checking

The pH and Bromine levels must be checked between animals. The pH must be adjusted first followed by the Bromine level.

Following a few days of testing after each treatment, the dosing of the pH and Bromine will gradually be such that the testing times can be extended.

Minimum frequency for checking should be daily.

5.8 Chlorine systems

5.8.1 Adjustment

a) Maintain Chlorine levels between **1** and **3 ppm**.

b) With fresh water add **1/5 ounce** to a full storage tank. This should give a reading of **2 ppm**.

c) To maintain Chlorine level, add **1/12 ounce** to a full storage tank. This should increase the reading by **1 ppm**.

5.8.2 Checking

The pH level and Chlorine levels must be checked between animals. The pH must be adjusted first followed by the Chlorine or Bromine level.

Following a few days of testing after each treatment, the dosing of the pH and Chlorine/Bromine will gradually be such that the testing times can be extended.

Minimum frequency for checking should be daily.

5.9 Microbial tests

Routine tests must be made for:

The total bacterial count and for coliforms.

- E.coli
- Pseudomonas aeruginosa
- Legionella

Testing for Cryptosporidium and Giardia is not required, although there is a residual risk of infection from these pathogens. A risk evaluation must be made by the responsible person to assess whether testing for these pathogens is necessary for their particular circumstances.

Testing for Staphylococcus aureus is not required, although there is a residual risk of infection from this pathogen. A risk evaluation must be made by the responsible person to assess whether testing for this pathogen is necessary for their particular circumstances.

5.9.1 Frequency of microbial testing

For hydrotherapy use testing twice a week is required for the total bacterial count, coliforms, E.coli and Pseudomonas aeruginosa. Three monthly testing is required for Legionella.

NOTE: A risk assessment must be made by the person responsible for operating the HydroWorx system to assess that this recommended frequency of testing is suitable for their particular circumstances. The testing frequencies may need to be adjusted up or down according to their assessment and their experience using the system.

5.10 Spares/Parts list

Part no	Description	Quantity Per system
6/284	Door/Window seal	4.5m
35/CBP0555	Treadmill belt	1
50/116	Water level sensor	1
50/705	Actuator	1
21/CR0237	Tail Roller	1
465/0040	Tail Roller, Graphalloy bush	2
21/CR0236	Drive drum	1
465/0221	Circulation pump - **Certikin NEWMP033M**	1
465/0208	Transfer pump - **DAB EUROSWM 150M 230V 1PH 50Hz**	2
452/1166	Sand filter brazil 15" side mount	1
452/0015	Filter media – 20kg bag	2
452/0096	2 KW elecro flowline S/S heater	1
465/0219	Certikin uv clarifier CUV-15V4	1
552/GTK121B	Bromine/chlorine test strips	1
45/CHOG5066	pH plus	1
45/CHOG5065	pH minus	1

5.11 USA Spares/Parts list

Part no	Description	Quantity Per system
465/0248	Circulation pump	1
465/0226	Transfer pump - **DAB EUROSWM 150M 230V 1PH 60Hz**	1
452/H001	1.5KW coates ils ser. heater	1

5.12 Spare parts

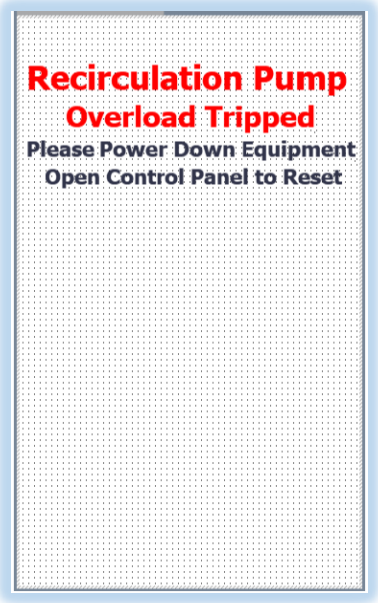
Contact the HydroWorx service team for spares and replacement parts.

6. TROUBLESHOOTING

6.1 Fault information messages

Fault information messages are displayed on the K900 touch screen with the cause and remedial action required.

Below is an example error message.



In the event of an error, first consideration should be to animal safety.

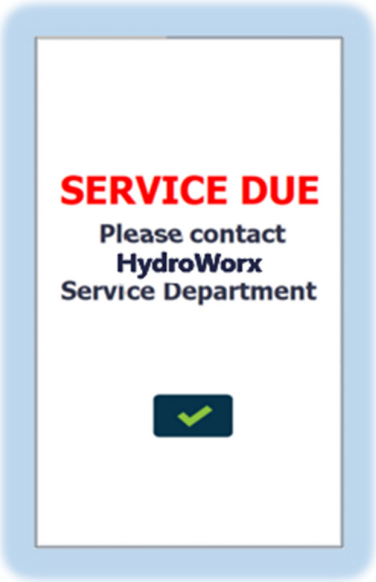
If there is a animal in the chamber assist them out before attempting to fix the error with the equipment.

6.2 Troubleshooting

Fault	Screen message	Action
No power when switched on	None	Check, Mains power distribution board. Correct power supply specification.
Door leak	None	Check: Clean seal. Adjust door brackets. Replace seal.
Glass leak	None	Replace/ Refit glass seal.
Noise from treadmill	None	Check, Guarding fitted correctly and secure. Track treadmill belt.
No display on touch screen	None	Check, Screen MCB.

6.3 Servicing information

To keep the system operating efficiently it should be serviced by HydroWorx Service after 1500 hours of use or 12 months whichever is the sooner.
When the service is due, at the start of each session a service reminder will show.
Continue to use the K900 by confirming this message.



7. INSTALLATION & COMMISSIONING

Installation and commissioning of K900 equipment should only be carried out by HydroWorx personnel or personnel trained by HydroWorx.

Refer to the sales manual, pre-installation guide and/or associated drawings for the site layout and conditions for the particular system supplied.

NOTE: That the treadmill chamber must not be sited in an area where hard, sharp or metallic objects are expected to be present or for instance, in a corridor.

Refer also to the routine inspection & maintenance section of this manual for all procedures relating to filling of the system and water treatment chemicals etc.

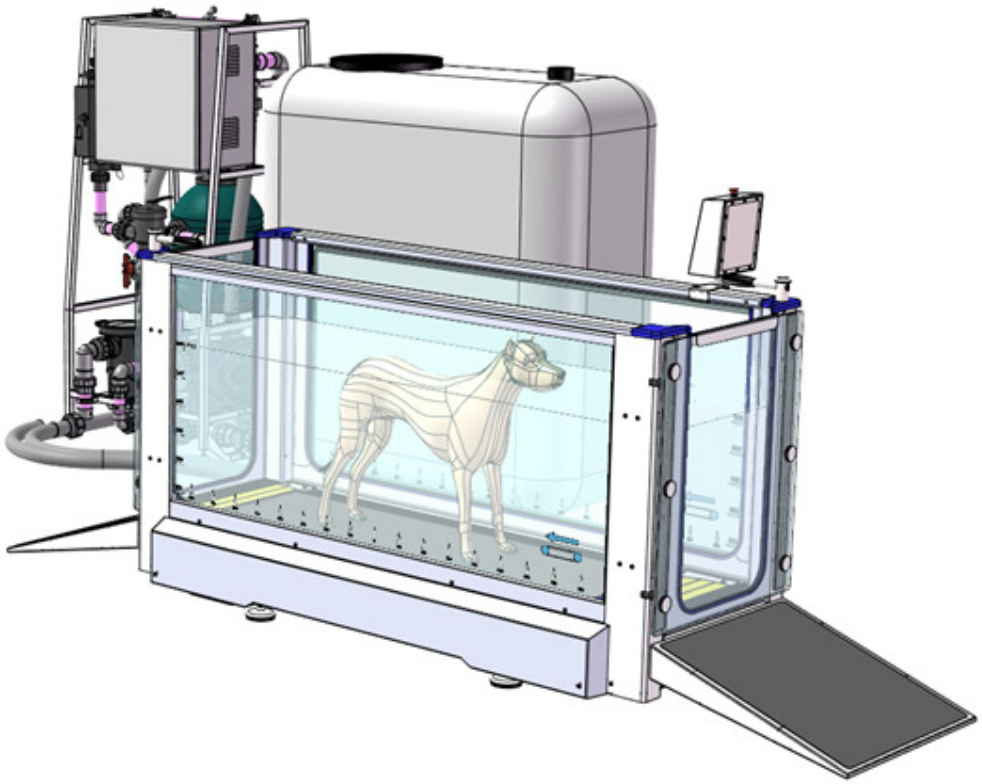


Figure 7-1

7.1 Transportation and packaging

7.1.1 Removal of packaging

All packaging should be carefully removed from the components.

The treadmill chamber can be moved using a pallet truck, ensure transportation castors if fitted are removed from the treadmill chamber before use.

NOTE: At all times great care should be taken when unpacking and moving the treadmill chamber to avoid damaging the glass.

7.1.2 Disposal of packaging

The packaging requires no special method of disposal.

7.2 Safety information

7.2.1 Basic precautions when using lifting equipment to handle the K900

Never stand or walk underneath the K900 equipment when it is lifted.
Always stay outside of the area where the K900 could fall if a lifting accessory fails.
Only use slings and lifting accessories with a load rating greater than the weight to be lifted, and which have been tested in accordance with applicable regulations for lifting equipment.

7.2.2 Working at height

The following transport and installation activities expose personnel to hazards of falling:

- Working on the lorry deck when unloading the K900.
- Retrieving slings and lifting accessories after lifting K900 into place.
- Assembling of K900 components at height.

7.2.3 Manual handling

The following transport and installation activities expose personnel to hazards relating to manually handling heavy objects:

- manual lifting of the K900 components.
- manually pushing and pulling the K900 on castors to the point of installation, especially on sloping floors.

7.2.4 PPE used during transportation and installation

PPE can reduce the severity of injuries from certain hazards.

7.2.4.1 PPE Recommendations

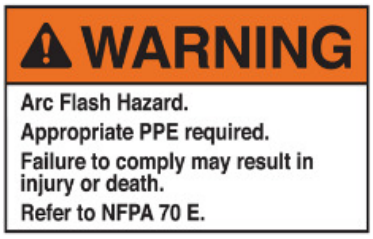
Protection	PPE	Hazard	Hazardous Event & Activity
Head	Hard hat	Impact/crushing	Unexpected horizontal movement of machine when lifting Components or tools falling as the machine is raised
Hands	Gloves	Cuts/abrasions	Sharp edges and corners when manually handling machine Chemical mortar used for floor fixings
Body	Overalls	Cuts/abrasions	Manual handling generally
Eyes	Safety glasses	Debris	Drilling floor for fixings
Ears	Ear defenders	Noise	Drilling floor for fixings
Feet	Safety footwear	Crushing	Manually handling/transporting the machine

7.3 Plant pack and water storage installation

Refer to the sales manual, pre-installation guide and associated drawings for the site layout and conditions for the particular system supplied. Also see the maintenance section of this instruction manual for all procedures relating to filling of the system and water treatment chemicals etc.

7.3.1 Power supply

USA: 220Vac, 25 Amp, 60Hz single phase, with 5mA RCD protection.



Grounding must be in compliance with local legislation.

7.3.2 Water supply

The storage tank holds 396 US gallons. Refer to the procedures in Section 5.6.6, Testing and maintaining chemical levels for maintaining the K900 unit.

A fresh water supply should be provided to fill the water storage tank.

A suitable drainage facility should be provided locally to the equipment. Waste water is dumped via a hose from the plant pack.

Flow rate 64 US gallons/minute.

7.3.3 Waste water drain

All waste water is removed from the system via the sand filter and the manual dump valve. See the maintenance section of the instruction manual for details of the sand filter operation. It is recommended that the end of the pipe from the waste port on the sand filter and dump valve be secured in place over the drain.

The waste water can be disposed of down a standard “foul” drain without any prior treatment. The waste drain must be sized to accept the maximum flow rate from the system, which is 64 US gallons/minute.

7.3.4 Siting the water storage tank

Water storage tank foot print: 61” x 29”.

The water storage tank should be sited on firm level ground. The maximum floor loading is 3463 lb including water.

The location should allow access for service pipes to one end of tank.

It is recommended that the water storage tank is located close to the plant pack.

The maximum distance from the plant pack is 32 feet.

7.4 Siting the plant pack

Plant pack footprint: 42”x29”

The plant pack should be sited on firm level ground, maximum floor loading 430 lb including the main control panel.

The location should allow access for service hoses to both sides of the plant pack and the front of main control panel should be accessible. Electrical conduit and water pipes run from the plant pack to both the treadmill chamber and the water storage tank. The maximum distance from the treadmill chamber and the Plant Pack is 32 feet.

The operator panel should be mounted on the treadmill chamber and connected via an electrical conduit to the main control panel on the plant pack.

7.4.1 Electrical connections to the plant pack

The cables running from the plant pack to the main control panel are to be connected as shown on the electrical schematic provided at the end of this manual.

7.4.2 Other electrical equipment in the vicinity of the plant pack

Any other electrical equipment in the plant pack room must be rated IP44, in other words protected from the ingress of splashing water from any direction.

7.4.3 Water connections to the plant pack

The flexible pipes are connected from the plant pack to the exercise chamber as shown in the hydraulic circuit at the end of this manual.

7.5 Exercise chamber installation

Refer to the sales manual, pre-installation guide and associated drawings for the site layout and conditions for the particular system supplied.

NOTE: That the exercise chamber must not be sited in an area where hard, sharp or metallic objects are expected to be present or, for instance, in a corridor or in close proximity to gym equipment.

7.6 Siting the treadmill chamber

Treadmill Chamber foot print:135” x 33” including access ramps.

The treadmill chamber should be sited on firm level ground. The maximum floor loading is 3300 lb, including water at maximum fill level. The floor surface should be suitable for a small amount of water spillage and have a non-slip finish.

The ideal chamber location should give approximately 3 feet of access all around the chamber for operator and animal. The treadmill chamber must be set level using the adjustable feet.

Consideration should be given to the possibility of accidental release of water from the chamber.

Locating electrical equipment in the area of the chamber should be avoided.

The treadmill chamber can be sited on an upper floor level to the plant pack and water storage tank but not below. The plant pack and water storage tank should always be sited on the same floor level.

7.7 Water drain

Floor drainage in the treadmill chamber room is required. The specification of the floor drain is as given in the pre-installation guide.

7.8 Other electrical equipment in the vicinity of the exercise chamber

The exercise chamber room should be treated as a bathroom with regards to electrical safety. Compliance with the requirements of the “locations containing a bath or shower” section of the IEE Wiring Regulations must be maintained.

7.9 Ventilation

The working environment of the exercise chamber must have sufficient ventilation. Consider and assess the air flow, room temperature and humidity.

7.10 Cleaning

The chamber should be cleaned before use as specified in Section 4, Cleaning.

8. DECOMMISSIONING & DISPOSAL

8.1 Lifespan of the K900

There is no definite maximum lifespan for the K900 system. The examination schedule should be reviewed to take into consideration the age of the machine. Inspections should take into account the possibility of deterioration in welded joints and fatigue, in addition to the routine inspection and maintenance schedule.

Although the Machinery Directive does not require HydroWorx to retain the technical records after a period of 10 years, we intend to maintain these records indefinitely. However, provision should be made for the possible lack of technical support after 10 years.

8.1.1 Estimated lifespans

Sub-Assembly	**Maximum estimated lifespan**
The chamber, including any electrical equipment fixed to it & the treadmill conveyor.	10 years
The water tank & pump system including all the pipework.	10 years
The electrical control system, including all cabling to the other sub-assemblies.	10 years

If component parts of a sub-assembly are replaced during the estimated lifetime, the sub-assembly must still be disposed of at the end of its lifetime. If a sub-assembly is replaced entirely, then the estimated lifetime of that sub-assembly starts from the date of its installation.

8.2 Decommissioning procedure

a) Drain the system.

b) Isolate the electrical supply at the wall socket.

c) Remove all other electrical equipment from the rooms. No mains voltage equipment such as power tool heaters or lighting must be brought into the plant pack or chamber rooms while water is being drained from the system.

d) Disconnect the pipes from the exercise chamber.

e) Drain the water in the pipes and in the plant pack pipework.

f) Separate the system into its component sub-assemblies.

8.3 Disposal

The K900 system can be returned to HydroWorx for disposal. Alternatively, it can be disposed of in accordance with the regulations in force at the time and place of disposal.